



A Federated Learning Approach for Reliable Diabetes Prediction

Comparative Analysis of Machine Learning and Deep
Learning Models with Probability Calibration

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Abstract

Background: Credit scoring is a mechanism used by lenders and financial organizations to determine the creditworthiness of individuals or businesses requesting credit or loans. It involves assessing various indicators to determine the likelihood of a borrower repaying their loans on time. The primary goal of credit scoring is to evaluate the risk involved in lending money to a particular borrower. Additionally, rule-based machine learning algorithms are employed to determine whether an individual is eligible to receive a loan.

Objectives: The thesis compared the RIPPER and CN2 algorithms to determine their usefulness in predicting personal loan credit scores. The primary objective was to assess model performance using accuracy, precision, recall, and F1 score metrics. The study aimed to determine which of these two rule-based machine learning algorithms gave more accurate and dependable predictions in the context of personal loan credit scores.

Method: The thesis used an experimental methodology to achieve its objectives and answer the research questions. The Credit Score Classification dataset from Kaggle is chosen, which is extremely helpful since it provides critical data to determine creditworthiness, which is an important part of the banking business. Using this method, a comparison was made between two rule-based machine learning (ML) algorithms, RIPPER and CN2. The experimentation step involved rigorous training, validation, and testing of these algorithms on the chosen dataset. Performance evaluation was then carried out, concentrating on key measures such as accuracy, precision, recall, and F1 score. This allowed for a complete comparison of the algorithm's performance in predicting credit scores for personal loans. Hyperparameter refining optimized model performance through extensive testing with different parameter values. Model validation was carried out using k-fold cross-validation. Finally, the study aimed to find the superior algorithm based on the utilized performance metrics.

Results: According to the experiment results, CN2 outperformed the two chosen machine learning algorithms, with an accuracy rate of 47.51% against RIPPER's 38.24%. The precision analysis found that CN2 has a somewhat greater precision of 42%

Conclusions: CN2 outperformed RIPPER in accuracy, precision, and F1 score, indicating its ability to make accurate predictions when identifying creditworthy loan applicants. While RIPPER had a slightly higher recall, suggesting its effectiveness in identifying positive cases, CN2 achieved a better balance between precision and recall. CN2's more balanced and consistent performance across all measures makes it the preferred option for personal loan credit scoring applications.

Keywords: Machine Learning, RIPPER, CN2, Rule-Based Models, Credit Scoring.

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Diabetes mellitus is one of the most important chronic health concerns of today's world, impacting about 537 million individuals worldwide and expected to reach 783 million by 2045 if current trends continue. The disease occurs primarily in two forms: Type 1 diabetes, in which the immune system destroys insulin-producing cells, and Type 2 diabetes, in which the body becomes increasingly resistant to insulin. Type 2 diabetes accounts for 90-95 percent of all occurrences and is closely associated with changeable lifestyle variables such as obesity, physical inactivity, and poor eating habits. If left untreated, it gradually destroys organs over time, eventually leading to serious problems such as cardiovascular disease, renal failure, nerve damage, and eyesight loss. Because many people go unnoticed for long periods of time, early and accurate diabetes risk identification is more than a clinical convenience; it is a medical need that may significantly change the course of the disease and avert irreversible harm (Acharya et al. 2008).

Diabetes diagnosis requires the simultaneous analysis of multiple clinical and demographic variables, including fasting glucose concentration, body mass index, blood pressure, insulin levels, and age, making manual decision-making more difficult for healthcare professionals as patient data volumes increase. Machine learning has emerged as an effective method for assisting with early diagnosis by automatically understanding complex correlations between these variables and diabetes outcomes. Classical models such as Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting have shown strong and interpretable performance on structured diabetes datasets (Polat and Güneş, 2007; Acharya et al., 2008), whereas deep learning architectures such as Multilayer Perceptrons have gained popularity for capturing non-linear risk factor relationships with competitive predictive accuracy (Smith et al., 1988; Mohamed et al., 2002). Despite these advancements, the great majority of published diabetes prediction models rely on centralized data collecting, which is not only technically convenient but also illegal in most real-world healthcare deployment circumstances.

- *The Privacy Barrier in Healthcare Data*

Under strong data protection laws, healthcare organizations around the world are prevented from sharing patient data beyond organisational boundaries. In Europe, the General Data Protection Regulation establishes rigorous criteria for the authorized processing of personal health data, including stringent prohibitions on cross-institutional data transfers. In the United States, the Health Insurance Portability

and Accountability Act establishes federal standards for safeguarding individually identifiable health information. India’s Digital Personal Data Protection Act 2023 provides similar safeguards to one of the world’s largest and most diversified patient populations (Naithani, 2024; Tschider et al., 2024). These frameworks collectively imply that patient records stored by various hospitals, clinics, and regional health centres cannot be lawfully combined into a single centralized dataset, even if doing so would result in more accurate and generalizable prediction models.

This generates a basic conflict at the core of healthcare AI. The data required to build effective, generalisable diabetes prediction models is spread among hundreds of institutions, each with a limited and potentially biased view of the patient population. Machine learning models trained on constrained single-institution datasets usually have low generalizability and might include demographic biases specific to that institution, decreasing their effectiveness and fairness when used in larger clinical contexts (Schmidt et al., 2018). The legislative system that safeguards patient privacy also restricts large-scale data collaboration, which would enable AI-powered diabetes prediction genuinely reliable on a population scale. Resolving this conflict is the primary motivation for the federated learning approach used in this research.

• *Federated Learning as a Solution*

Federated learning, presented by McMahan et al. (2017) via the Federated Averaging method, provides a principled solution to the privacy-performance issue. Rather than centralizing patient records, federated learning allows many institutions, known as clients, to collaborate on training a shared global model by exchanging just model parameters rather than raw data. Each client trains a local model on its own private dataset, sends the weights to a central aggregation server, and receives an updated global model in exchange. This iterative procedure is repeated across numerous communication cycles until the global model converges, with no patient records leaving the source institution during the entire training phase.

Over the last five years, the healthcare research community has become increasingly enthusiastic about this model. Nguyen et al. (2022) conducted a comprehensive review of over 150 federated healthcare research and documented its successful application in health data management, remote patient monitoring, medical imaging, and disease prediction. Pati et al. (2024), in a Cell Patterns analysis of privacy protection in federated healthcare AI, confirmed that federated learning significantly decreases privacy risks while enabling collaborative model improvement across institutional boundaries. Applied especially to diabetes prediction, Islam et al. (2022) demonstrated that federated models trained on real clinical electronic health record data from different healthcare organisations can reach performance comparable to centralised training while fully protecting patient privacy. Tang et al. (2024) applied this conclusion to real cross-provincial primary care data from Canada, demonstrating federated feasibility in a realistic multi-site context with no patient data leaving the originating province. Su et al. (2023) demonstrated that secure federated protocols for diabetes risk prediction may prevent gradient leakage attacks while retaining

competitive accuracy. These works collectively demonstrate that federated learning is technically viable for diabetes prediction; nevertheless, they also have a major common drawback, which this thesis addresses directly.

- *The Trustworthiness Problem*

Every existing federated diabetes prediction study assesses its models primarily using classification accuracy metrics including accuracy, F1-score, and AUC. While these metrics assess whether a model predicts the correct diagnosis, they do not address whether the model’s estimated probability are reliable, if its reasoning is understandable to doctors, or whether it treats different patient groups fairly. These three qualities — probability dependability, explainability, and fairness — are exactly what helps decide whether a model can be trusted and used in a clinical setting, yet they are consistently absent from the existing federated diabetes research.

In terms of probability reliability, clinical choices for diabetes treatment are made using predicted risk scores rather than binary labels. A physician’s decision to start preventive treatment is based on whether a predicted 75 percent risk actually means 75 percent, not 55 percent or 90 percent. This is known as calibration, and Silva Filho et al. (2023), in the definitive survey on classifier calibration, demonstrated that modern machine learning classifiers such as Random Forest and Gradient Boosting are systematically overconfident, producing probability estimates that cluster near the extremes instead of reflecting true outcome frequencies. In healthcare applications where prediction confidence is as important as prediction accuracy, overconfidence or incorrect probability estimations can result in hazardous clinical interventions or missed early diagnosis (Jagatheesaperumal et al., 2024). Despite this, every federated diabetic study analyzed in this thesis — Islam et al. (2022), Tang et al. (2024), Su et al. (2023), Gawande et al. (2025), Chellamani et al. (2025), Yang and Li (2025), and Hasan and Li (2026) — completely ignores probability calibration.

On the explainability dimension, medical practitioners must first comprehend and trust AI-generated predictions before acting on them. Black-box models that provide accurate predictions but lack interpretable logic undermine clinician confidence and pose significant challenges to real-world implementation (Awosika et al., 2024). Regulatory frameworks include GDPR. Article 22 provides a legal foundation for the right to explanation in automated decision-making systems, making explainability a legal obligation rather than a desired feature in European healthcare settings. SHAP, or Shapley Additive Explanations, has emerged as the main post-hoc explainability method, offering theoretically based per-feature contribution ratings that can be interpreted at both the local instance and global population levels. In a federated context, SHAP must be computed locally at each client, which raises another concern raised by Ducange et al. (2024): explanation consistency may vary across clients when data distributions differ under non-IID conditions, potentially producing contradictory clinical guidance from the same global model at different institutions. Among existing federated diabetes studies, only Gawande et al. (2025) uses an explainability technique, LIME rather than SHAP, and does not measure explanation

consistency between clients or rounds.

In terms of fairness, a model that performs well on aggregate measures may behave substantially differently across demographic subgroups defined by age, BMI, or gender. Equalized odds were defined as the recommended fairness metric for clinical prediction tasks by Xu et al. (2022), and Chen et al. (2023) observed that federated learning can serve as a bias reduction method by allowing training across demographically varied institutional datasets. Despite this, no extant federated diabetes study assesses demographic parity or equalized chances across any subgroup, creating a gap in clinical readiness assessments for all published approaches.

- *Identified Research Gap*

The absence of probability calibration, explainability, and fairness evaluations in federated diabetes research creates a considerable and significant gap between technical performance and clinical availability. Some studies look into explainable AI, but only in centralised systems — Alam et al. (2025) used SHAP to predict diabetes without a federated component, and Tasin et al. (2023) demonstrated strong SHAP and LIME explanations on the Pima Indians dataset without considering privacy preservation. Others focus solely on federated learning, discarding explainability and calibration completely. Kumar et al. (2025) provide the closest existing integration, combining federated learning, SHAP, and isotonic calibration for liver disease prediction — but their work addresses a different disease domain, does not compare classical ML models to deep learning under federated calibration, does not assess explanation stability across communication rounds, and does not include a fairness analysis. In diabetes prediction, no study has yet coupled federated learning with probability calibration, SHAP-based explainability, and fairness auditing in a single unified evaluation framework across both IID and non-IID data distribution scenarios.

Another weakness recognized by the existing literature is the constant lack of communication cost analysis in federated diabetic studies. Communication overhead was identified as one of the five unresolved challenges in healthcare federated learning by Nguyen et al. (2022), and McMahan et al. (2017) acknowledged that while FedAvg reduces communication rounds when compared to synchronous gradient descent, the cumulative bytes transmitted across all rounds represent a genuine deployment cost that grows with model size, client count, and the number of rounds to convergence. There is no previous federated diabetes study that reports this cost, making it hard to determine the practical communication cost required for deployment.

- *Contributions*

This thesis addresses all five identified gaps by proposing and evaluating TrustFed Diabetes - a federated learning framework for diabetes prediction that evaluates trustworthiness across five dimensions simultaneously. The framework trains five classical machine learning models — Logistic Regression, Random Forest, Gradient Boosting

Classifier, Decision Tree, and XGBoost alongside a shallow Multilayer Perceptron under FedAvg and FedProx aggregation, with isotonic regression and sigmoid calibration applied locally at each client, SHAP computed independently at each client to preserve data locality, and all results evaluated across IID and non-IID conditions using Dirichlet partitioning at three heterogeneity levels on the CDC Diabetes Health Indicators dataset containing 253,680 records. The five evaluation dimensions are predictive performance, probability calibration quality, SHAP explanation stability across federated rounds, subgroup fairness across age and BMI categories, and total communication cost to convergence.

1.1 Ethical, societal and sustainability aspects

- **Ethical Aspects:** This thesis uses sensitive medical data for the prediction of diabetes, which poses significant ethical considerations, particularly about patient privacy, confidentiality, and data security. Medical data cannot be freely shared due to legal and ethical constraints, hence this study employs a federated learning strategy. In this configuration, patient data is kept inside local healthcare institutions such as hospitals or clinics and never moved to a central server. Instead, only model updates (learned patterns) are exchanged between institutions, thereby protecting patient identity and ensuring compliance with data protection standards such as GDPR and HIPAA.

Although data is not stored centrally, the system still enables collaborative learning across many healthcare facilities, making it both privacy-preserving and effective for real-world medical applications. In addition, explainable AI approaches are used into the model to increase transparency. This lets clinicians to understand why a prediction (such as diabetes risk) is produced, rather than viewing the model as a "black box." This builds trust and promotes better clinical decision-making. Furthermore, probability calibration is used to guarantee that the predicted probabilities are accurate and realistic. This reduces the risk of making overconfident or misleading forecasts, which is especially significant in healthcare, where inaccurate confidence levels might influence treatment decisions. Overall, the combination of federated learning, explainability, and calibration leads to a more secure, transparent, and reliable diabetes prediction system.

- **Societal considerations:** From a societal perspective, this recommended approach encourages early detection of diabetes, which can help decrease major long-term health problems and improve overall patient outcomes. Using a privacy-preserving federated learning system, healthcare institutions can collaborate without revealing sensitive patient data. This enables the development of more robust and generalizable models that can help both major hospitals with advanced resources and smaller or resource-limited healthcare organizations. Furthermore, the study takes into account real-world variances in patient

data between hospitals (known as non-IID data). By evaluating the model under these settings, the system improves its ability to handle population diversity while reducing potential bias. This ensures that projections are fair and reliable across patient groups, resulting in more equitable healthcare decision-making.

- **Sustainability Aspects:** The thesis emphasizes sustainability by using efficient classical machine learning models and a shallow neural network, which need less computational resources than big deep learning architectures. Federated learning promotes sustainability by minimizing the requirement for centralized data storage and avoiding the continual transmission of raw sensitive information to a central server. However, it adds communication cost due to iterative model updates between clients and servers. Despite this, the decentralized nature of federated learning enables scalable and privacy-preserving deployment of AI systems in healthcare, making it ideal for long-term use with less reliance on centralized infrastructure and data storage systems

1.2 Problem Definition

The rising global prevalence of diabetes has made early and accurate diagnosis a major concern in modern healthcare. However, developing successful AI-driven prediction systems in healthcare settings presents major hurdles that extend beyond obtaining high accuracy. One of the most serious concerns is data privacy. Patient health records are extremely sensitive, and healthcare organizations are legally and ethically required to keep them secure. Traditional machine learning algorithms require centralizing data from various sources to train a model, which raises major privacy hazards and frequently violates rules such as GDPR and HIPAA. As a result, many institutions are hesitant to share their data, limiting the amount and variety of data accessible for training viable models.

Even when models are created, they frequently behave as black boxes, making predictions but providing no explanation for why a certain decision was made. This is a significant issue in the clinical setting. A doctor cannot act properly on a forecast that they do not understand, and patients need to know how decisions concerning their health are made. The lack of interpretability reduces trust in AI systems, limiting their use in real-world healthcare situations.

Beyond interpretability, there is another subtle but crucial issue: inaccurate probability outputs. Many machine learning models generate predicted probabilities that do not adequately reflect a patient’s actual likelihood of having diabetes. For example, a model may assign a 90% risk score while the true probability is much lower, prompting physicians to make incorrect decisions. This miscalibration can have major effects in medical settings where risk assessments have a direct impact on treatment planning. Furthermore, real-world healthcare data is rarely consistent. Different hospitals serve distinct patient populations, resulting in non-uniform or

non-IID data distributions among institutions. Most existing research evaluate models under ideal, balanced data conditions, which do not reflect the messy reality of clinical settings.

This thesis addresses all of these interconnected difficulties by introducing a federated learning architecture that enables various healthcare institutions to train diabetes prediction models together without ever exchanging raw patient data. The goal of integrating explainable AI approaches and probability calibration methods is to create a system that is not only accurate, but also one that physicians can trust, comprehend, and confidently employ in real-world choices.

1.3 Aim and objectives

1.3.1 Aim

The purpose of this thesis is to build and construct a federated learning framework for diabetes prediction that will allow multiple healthcare institutions to train models collaboratively while maintaining patient data privacy. The study compares the performance of classical machine learning models and deep learning approaches in both IID and non-IID data sets, while also incorporating explainable AI techniques to improve model transparency and probability calibration methods to ensure that predicted risk scores are statistically reliable and clinically meaningful.

1.3.2 Objectives

- To design and implement a federated learning framework that protects privacy so that multiple healthcare organizations can work together to develop diabetes prediction models without exchanging private patient information.
- To create and evaluate a shallow Multilayer Perceptron and traditional machine learning models for diabetes risk classification in both IID and non-IID data distribution scenarios.
- To incorporate explainable AI methods based on SHAP to improve model transparency and guarantee prediction stability throughout multiple training rounds.
- To assess model fairness across demographic subgroups using demographic parity and equalized odds metrics, and to increase the accuracy of predicted risk probabilities using isotonic regression and sigmoid based probability calibration.
- To evaluate the federated framework’s communication efficiency and provide a unified system that is accurate, understandable, calibrated, fair, equitable, and practically implementable in actual clinical settings.

1.4 Research questions

RQ1. How closely does federated learning match centralised diabetes prediction performance across IID and non-IID data distributions in terms of AUROC and F1, what percentage drop in performance is acceptable for clinical deployment, and what cumulative communication cost in bytes is required to reach stable convergence?

Motivation: Early and accurate diabetes prediction requires huge amounts of patient data from numerous healthcare institutions, but data security rules such as GDPR, HIPAA, and India’s DPDP 2023 explicitly prohibit centralized data exchange across organizational borders. Federated learning addresses this by allowing collaborative model training via model parameter sharing rather than raw patient data transfer, which protects privacy while allowing institutions to benefit from each other’s data. However, the practical viability of federated learning for clinical deployment is dependent on three unanswered questions: how closely its predictive performance matches that of a centralised model trained on combined data, what level of performance drop in terms of AUROC and F1 is acceptable without compromising clinical utility, and how much communication cost in cumulative bytes is required to achieve stable model convergence. Real hospital data is often consistently distributed across institutions; different sites serve diverse patient demographics, resulting in non-IID data situations that can reduce federated model performance when compared to optimal IID settings. Before proposing federated learning as a clinically deployable alternative to centralised diabetes prediction, it is important to understand whether it maintains competitive performance under both IID and non-IID settings, as well as the communication cost.

RQ2. How do isotonic regression probability calibration and SHAP-based explainability improve the quantitative trustworthiness of federated diabetes prediction models, measured through Expected Calibration Error, Brier Score, Log Loss, SHAP feature importance stability across federated rounds, and equalized odds difference across age and BMI subgroups?

Motivation: Accurate diabetes prediction alone is insufficient for clinical deployment since healthcare professionals must trust the model’s results in order to act on them. A model that predicts high diabetes risk with systematically overconfident probabilities may result in unnecessary healthcare interventions, whereas a model whose feature explanations shift unexpectedly between several training rounds cannot be reliably evaluated by clinicians. Furthermore, a model that performs well overall but acts differently across age and BMI subgroups presents major fairness concerns in a clinical context where equal treatment for all patient groups is essential. Despite these issues, no previous federated diabetes study has examined probability reliability using calibration measures, explanation consistency across communication rounds, and subgroup fairness inequalities in a single evaluation framework. This research question fills that gap by quantifying how isotonic regression calibration improves probability trustworthiness via ECE, Brier Score, and Log Loss, how SHAP-based explanations remain stable across federated rounds, and whether the federated framework produces equitable predictions across age and BMI subgroups as measured by equalized odds difference.

1.5 Scope

This thesis focuses on developing and testing a federated learning framework for diabetes prediction using the CDC Diabetes Health Indicators dataset, which is a publicly available structured clinical dataset of 253,680 records with 21 features covering health indicators such as blood pressure, BMI, age, physical activity, and general health status. The experiments are conducted in a simulated federated environment with 3 and 5 clients representing healthcare institutions. Data is partitioned under both IID and non-IID conditions using a Dirichlet distribution at concentration levels $\alpha = 0.1, 0.5, \text{ and } 1.0$ to simulate realistic inter-institutional data heterogeneity.

The study compares five conventional machine learning models—Logistic Regression, Random Forest, Gradient Boosting Classifier, Decision Tree, and XGBoost—with a shallow Multilayer Perceptron as the deep learning baseline. Two federated aggregating methods, FedAvg and FedProx, are compared across all setups. To improve the dependability of estimated risk probabilities, probability calibration is used at the client level, including isotonic regression and Platt scaling. SHAP is employed for post-hoc explainability, with each client computing both local instance-level and global feature-level attributions independently in order to preserve data locality.

The evaluation framework simultaneously covers five dimensions: predictive performance measured through AUROC, F1, precision, recall, and accuracy; probability reliability measured through Expected Calibration Error, Brier Score, and Log Loss; explanation stability measured through SHAP feature importance rank correlation across federated communication rounds; and subgroup fairness measured through demographic parity difference and equalized odds difference across age and BMI.

This thesis does not include actual hospital facilities, live patient data, or clinician surveys. All results are based on controlled simulation tests using the CDC dataset. The scope is purposely confined to binary diabetes classification on structured tabular clinical data; it does not include multi-modal data, longitudinal patient records, continuous glucose monitoring, or other chronic disease prediction tasks. Advanced privacy-enhancing approaches such as differential privacy, secure multi-party computation, and homomorphic encryption are recognized as essential directions, but they fall outside the scope of this work. Similarly, big deep learning architectures beyond shallow MLPs, personalized federated learning techniques, and real-time federated deployment across physical healthcare networks are proposed as future research directions rather than being addressed in this thesis.

1.6 Thesis Outline

- **Chapter 1— Introduction**

The first chapter provides an introduction to the research topic by discussing the clinical importance of early diabetes prediction and the regulatory challenges that prevent centralized data sharing among healthcare institutions un-

der frameworks such as GDPR, HIPAA, and India’s DPDP 2023. It defines the central issue — that existing federated diabetes prediction frameworks focus solely on accuracy while ignoring probability reliability, explainability, and fairness — and presents the two research questions, their motivations, the study’s goals and objectives, and the scope of the work. The chapter finishes by explaining the study’s significance in the context of privacy-preserving clinical AI and reliable medical decision assistance.

- **Chapter 2– Background**

The theoretical framework and mathematical foundations required to comprehend the procedures and outcomes of this thesis are discussed in Chapter 2. The clinical background of diabetes mellitus, machine learning models for binary diabetes classification, such as Logistic Regression, Random Forest, Gradient Boosting, Decision Tree, XGBoost, and the shallow Multilayer Perceptron, the mathematical foundations of federated learning, such as FedAvg and FedProx algorithms, non-IID data simulation using Dirichlet partitioning, probability calibration theory, such as isotonic regression and sigmoid calibration, SHAP-based explainability, algorithmic fairness metrics, such as equalized odds and demographic parity, and communication cost measurement in federated systems, are all covered.

- **Chapter 3– Related Work**

Chapter 3 presents a systematic literature review of 40-50 peer-reviewed publications from 2020 to 2026 organized into four topical clusters: federated learning in healthcare and diabetes prediction, non-IID data heterogeneity and aggregation algorithms, probability calibration in clinical machine learning, and explainable AI in federated healthcare systems. Each cluster is closely analyzed to see what existing research contributes and what remains unclear. The chapter concludes with a comprehensive comparison table of all reviewed studies and a synthesis of the five specific knowledge gaps addressed by this thesis: the absence of calibration in federated diabetes studies, the lack of integrated FL and XAI and calibration frameworks, unmeasured SHAP stability across federated rounds, no ML versus DL comparison under federated calibration, and the lack of fairness evaluation in federated diabetes prediction.

- **Chapter 4– Methodology**

The research method is fully described in technical detail in Chapter 4. The CDC Diabetes Health Indicators dataset is presented along with its preprocessing, which includes class imbalance handling and feature scaling; the federated learning environment setup with 3 and 5 simulated clients; IID and non-IID data partitioning using Dirichlet distribution at $\alpha = 0.1, 0.5$, and 1.0 ; local model training and hyperparameter optimization at each client; probability calibration using isotonic regression and sigmoid calibration; SHAP explainability integration at the client level; federated aggregation using FedAvg and FedProx with softmax-weighted performance-based aggregation; and a five-dimensional evaluation framework covering predictive performance, calibration quality, explanation stability, subgroup fairness, and communication cost.

- **Chapter 5– Results and Analysis**

Chapter 5 provides and examines the research findings, which are arranged around five evaluation dimensions and five comparison analyses. It compares client against federated learning to determine the advantages of collaborative learning over isolated local training in terms of accuracy, F1-score, and ROC-AUC. It then compares centralised and federated training to assess the privacy-performance trade-off, looking at how closely federated training matches a pooled-data centralised baseline in terms of accuracy, F1-score, and ROC-AUC while maintaining total data localization. The calibrated versus uncalibrated comparison assesses the improvement in prediction reliability brought about by isotonic regression and sigmoid calibration using Brier Score, Expected Calibration Error, and Log Loss. The IID versus non-IID comparison assesses the federated framework’s robustness in the face of realistic hospital data heterogeneity at all three Dirichlet concentration levels, as well as its performance and calibration consistency. Finally, the machine learning versus deep learning comparison looks at the differences in predictive behavior, calibration quality, and SHAP-based interpretability between the five conventional ML models and the shallow MLP in federated training.

- **Chapter 6– Discussion**

In Chapter 6, the results are discussed in relation to the expectations set out in the literature study and how they address the two research questions. It discusses the implications of SHAP explanation stability across federated rounds for clinical trust in federated systems, contextualizes the fairness findings across age and BMI subgroups in relation to actual demographic differences in diabetes prevalence, understands the centralized versus federated performance gap against the acceptable threshold for clinical deployment established in RQ1, and explains the significance of the calibration improvement across all model types in terms of clinical probability reliability. The chapter considers the limitations of the simulated environment and the generalizability of outcomes outside of the CDC dataset, and it links each result to the five knowledge gaps noted in Chapter 3.

- **Chapter 7– Conclusions and Future Work**

Chapter 7 concludes the thesis by summarizing the answers to both research questions and stating the work’s main contributions as the first unified federated framework for diabetes prediction that simultaneously addresses privacy, calibration, explainability, fairness, and communication cost measurement. It recognizes the limitations of using a single public dataset in a simulated rather than real federated deployment, and proposes concrete directions for future research, including real multi-institutional deployment across physical hospitals, integration of differential privacy and secure multi-party computation, personalized federated learning strategies for strongly non-IID settings, extension to other chronic disease datasets, and longitudinal patient risk monitoring.

2.1 Diabetes Mellitus

Diabetes mellitus is a chronic condition that changes how the body processes blood sugar. When you eat, your blood glucose levels rise, and the pancreas responds by releasing insulin, a hormone that functions like a key, allowing cells to absorb glucose and use it for energy. In diabetes, this mechanism fails. Either the pancreas is not producing enough insulin, or the body's cells have developed resistance to its effects, or both. As a result, glucose accumulates in the bloodstream rather than being absorbed, causing long-term damage to organs throughout the body. Diabetes is especially dangerous for public health because it develops quietly. A person can have dangerously high blood glucose levels for years, if not decades, without experiencing any symptoms. When evident symptoms arise — increased thirst, frequent urination, exhaustion, hazy vision — significant problems such as cardiovascular disease, kidney failure, peripheral nerve damage, and vision loss may have already occurred (Acharya et al., 2008).

Why Prediction Matters

Early identification is not just medically beneficial; it may significantly change a patient's path. Pre-diabetic interventions, such as lifestyle changes or early medication, might delay or even prevent the onset of complete diabetes, lowering the risk of significant consequences (Polat and Güneş, 2007). The problem is to identify which people are at high risk before they exhibit symptoms. This is where machine learning comes in. A well-trained model can identify high-risk individuals far earlier than traditional screening methods by learning complex, non-linear relationships between clinical measurements such as blood glucose levels, BMI, blood pressure, age, and family history and known diabetes outcomes in large patient datasets (Smith et al., 1988). The issue, however, is that the patient data required to train robust models is kept within hospitals and clinics that are prohibited by law from sharing it. That structural challenge serves as the primary reason for everything in this thesis.

2.2 Binary Classification

Diabetes prediction is modeled as a binary classification task. Each patient is defined by a collection of clinical measures (referred to as a vector x), and the model's purpose is to assess the patient's likelihood of having diabetes. Formally, the model

learns a function f that relates clinical measurements to probabilities.

$$p = f(x) \in [0, 1]$$

A result of 0.9 indicates that the model believes there is a 90% probability this patient has diabetes. A number of 0.15 indicates that it believes there is just a 15% possibility.

Turning Probabilities Into Decisions

To make an actionable clinical decision whether to refer for more testing or not the likelihood must be transformed into a binary label. This is done using a decision threshold (t):

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq t \\ 0 & \text{if } p < t \end{cases}$$

where $\hat{y} = 1$ means the model predicts diabetes positive and $\hat{y} = 0$ means diabetes negative.

Choosing t is not a machine learning challenge, but rather a clinical one. A lower threshold identifies more cases (greater recall), but it also flags more healthy patients (lower precision). A higher threshold decreases false alerts but risks overlooking people who truly require care. These trade-offs will be handled differently in different medical situations, such as mass screening programs versus specialist referral pathways. The methods chapter goes into much information about this threshold selection problem.

2.2.1 Machine Learning Models

This thesis compares six model families. Each offers a unique mathematical technique for learning the relationship between clinical characteristics and diabetes probability, as well as specific strengths, shortcomings, and compatibility with federated training.

Logistic Regression

Logistic regression is one of the oldest and most commonly used classification algorithms in medicine. The idea is straightforward: each clinical indicator (blood pressure, glucose level, BMI, and so on) is assigned a weight that represents how strongly it influences the prediction towards or away from diabetes. The model calculates a weighted total of all feature values, and the sigmoid function converts this sum to a probability between 0 and 1:

$$p = \frac{1}{1 + e^{-(w^T x + b)}}$$

During training, the model adjusts its weights to minimise how inaccurate its probability estimations are throughout the training set. This is measured using the binary

cross-entropy loss, a penalty that rises significantly anytime the model is confidently incorrect.

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

Logistic regression is particularly well-suited to federated learning for practical reasons: the entire model consists of only one number per feature plus a bias term. Sending those numbers via a network is essentially cost-free (McMahan et al., 2017). Its fundamental disadvantage is that it believes the distinction between diabetic and non-diabetic patients can be shown as a straight line through the feature space, which is an approximation that ignores complicated feature interactions. The sklearn implementation is utilised in both centralised and local experiments. A unique NumPy implementation is created for federated experiments that allows for direct gradient computation in accordance with the federated averaging protocol.

Decision Tree

A decision tree learns by continuously breaking down the patient dataset into smaller groups based on feature thresholds. At each split, it asks: which attribute, and which threshold for that feature, results in the clearest differentiation of diabetes and non-diabetic patients? The quality of a split is measured by Gini impurity — how mixed the two groups are after the split:

$$\text{Gini}(S) = 1 - \sum_{c \in \{0,1\}} p_c^2$$

$$\Delta \text{Gini} = \text{Gini}(S) - \frac{|S_L|}{|S|} \text{Gini}(S_L) - \frac{|S_R|}{|S|} \text{Gini}(S_R)$$

The tree expands until each leaf has a nearly similar population of patients. The estimated chance of a new patient reaching a leaf is determined by the proportion of diabetes patients in that leaf during training.

Decision trees are quite interpretable; you can pinpoint exactly which feature values resulted in a prediction. However, they are known for overfitting, and when a leaf has a small number of training examples, the resulting probability estimate can be quite incorrect. This makes them a particularly interesting scenario for probability calibration, which is a major focus in this thesis.

Random Forest

Random Forest solves the overfitting problem of single decision trees by training a huge ensemble of trees, each based on a randomly picked bootstrap sample of the training data and only evaluating a random subset of features at each split. This creates deliberate variation among the trees. The final predicted probability is the average over all T trees:

$$p_{RF}(x) = \frac{1}{T} \sum_{t=1}^T p_t(x)$$

Averaging across multiple different trees significantly reduces variation when compared to a single tree, resulting in more stable and typically stronger predictions. Random Forest consistently performs well in medical datasets. However, it has a well-documented calibration problem: its probability estimates are artificially pushed to the extremes, causing the model to be overconfident. Post-hoc calibration is thus very critical for this model (Silva Filho et al., 2023; Odesola et al., 2025).

Gradient Boosting Classifier

Gradient Boosting creates trees consecutively rather than training them singly and then averaging them. The first tree is simply an initial guess. Each succeeding tree is trained particularly to repair the current ensemble’s flaws, focusing its learning capacity on the circumstances where the ensemble is making mistakes. The learning rate (η) *determines the step size for adding new trees to the ensemble.*

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

The final probability is generated by passing the collected score through the sigmoid function.

$$p(x) = \frac{1}{1 + e^{-F_M(x)}}$$

Gradient Boosting is among the most powerful models for structured tabular data. Like Random Forest, its probability estimates are overconfident and benefit greatly from calibration (Ojeda et al., 2023).

XGBoost

XGBoost (Extreme Gradient Boosting) improves standard gradient boosting in two significant ways. First, it adds explicit regularisation terms to the objective function, penalising model complexity to prevent overfitting.

$$\text{Obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

Second, it employs a second-order Taylor approximation of the loss function rather than only the first-order gradient, resulting in a more precise signal for selecting splits and setting leaf values.

XGBoost is only utilised in centralised training, where it consistently produces the greatest AUROC of all models. This makes it the major performance upper bound

throughout the thesis, serving as the benchmark against which federated techniques are measured.

Shallow Multilayer Perceptron

The multilayer perceptron is a feedforward neural network that may describe non-linear interactions between inputs and outputs by routing data via one or more hidden layers of neurones. The shallow MLP utilised here has only one hidden layer. In the federated NumPy implementation, the forward pass operates as follows:

$$\text{hidden} = \tanh(XW_1 + b_1)$$

$$p = \sigma(\text{hidden} \cdot W_2 + b_2)$$

The centralised scikit-learn version employs a ReLU activation in the hidden layer:

$$\text{hidden} = \text{ReLU}(XW_1 + b_1) = \max(0, XW_1 + b_1)$$

The model’s total number of trainable parameters is:

$$P = d \times h + 2h + 1$$

where h represents the number of hidden neurones and d represents the number of input characteristics. The communication overhead every federated cycle is directly determined by the overall model size, which is $P \times 8$ bytes, since each parameter is stored as a 64-bit floating point number that takes up 8 bytes.

The MLP is the thesis’s only neural network, and it serves as the deep learning baseline in the federated architecture. It is uniquely compatible with federated averaging because its parameters are continuous real-valued matrices that can be meaningfully averaged across clients, as opposed to tree-based models, whose branching architectures and threshold rules cannot be integrated by averaging (McMahan et al., 2017). Neural networks are also often overconfident in their probability outputs, making them an important calibration test case (Guo et al., 2017).

2.3 Data Preprocessing

Before any model is trained, raw input data is preprocessed in three standard processes to ensure that it can be reliably learned from.

Missing value imputation: Real clinical datasets frequently contain gaps a patient may not have had a specific test. Missing numeric values are substituted with the feature’s median derived from training data:

$$x_{\text{imputed}} = \text{median}(X_{\text{train column}})$$

The mean is preferred over the median because it is more resilient to outliers, which are common in clinical measures.

Feature Standardisation: Different clinical aspects exist on vastly different numerical scales - blood glucose levels can range from 50 to 500 mg/dL, while ages range from 20 to 90. Models that rely on distance or gradient calculations may behave strangely when features have vastly varying magnitudes. Standardising each feature to zero mean and unit variance puts them all on equal footing:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

One-hot encoding: Categorical data, such as a feature identifying gender or ethnicity, cannot be supplied into a model as raw category labels. One-hot encoding transforms each category into a binary indicator column.

A fundamental rule: All preprocessing statistics, such as the median for imputation and the mean and standard deviation for standardisation, are computed exclusively on the training data and applied unchanged to the calibration and test sets. Fitting the preprocessor to the entire dataset, including test data, might result in information leakage, inflating measured performance (Kaliappan et al., 2024).

2.4 Centralised Learning

Centralised learning is the most straightforward training paradigm to understand: all patient data from all institutions is pooled in a single location, and one model is trained on the entire combined dataset. Mathematically, it minimises one global loss:

$$w^* = \arg \min_w \frac{1}{n} \sum_{k=1}^K \sum_{i \in D_k} L(f_w(x_i), y_i)$$

This is the maximum performance limit; the model has simultaneous access to all training examples and learns the most extensive, generalisable parameters possible (Islam et al., 2022). Throughout this thesis, federated results are measured against things imposed by centralised training results.

Centralised training serves as a theoretical benchmark, representing the best performance that could be achieved if this were not prohibited by GDPR in Europe, HIPAA in the United States, and India’s Digital Personal Data Protection Act 2023 (Naithani, 2024; Tschider et al., 2024). The legal barrier is real and cannot be negotiated. Centralised training thus acts as a theoretical benchmark the best performance you could achieve if data could flow freely rather than anything usable in practice. Federated learning is specifically designed to address the gap between what is possible in theory and what is legally acceptable.

2.5 Local-Only Learning

Local-only learning represents the polar opposite. Each institution develops its own separate model using only its own data, sharing nothing with anyone else.

$$w_k^* = \arg \min_{w_k} \frac{1}{n_k} \sum_{i \in D_k} L(f_{w_k}(x_i), y_i)$$

From a privacy sense, this is perfect; no data or model parameters ever leave the institution. However, from a predictive performance perspective, the consequences can be severe. A tiny rural clinic with 200 patients and minimal population variety generates a model that may operate pretty well for the specific population it has seen but performs badly in general. A hospital that primarily serves one age group or ethnic group will create a model with systematic blind spots.

This thesis’s limit for privacy-preserving performance is local-only training. It addresses the most fundamental argument for federated learning: can federating actually help? The average performance of all K local-only models is calculated as follows:

$$\text{Avg Local-Only} = \frac{1}{K} \sum_{k=1}^K \text{score}_k$$

If federated learning cannot consistently exceed this average, the overall communication overhead of federation is difficult to justify (Li et al., 2022).

2.6 Federated Learning

McMahan et al. (2017) presented federated learning, which answers the basic problem between wanting to train on big multi-institutional datasets but being legally banned from centralising that data. The fundamental discovery is elegant: while patient records cannot be shared, the mathematical weights that a model learns from them do not include any patient information and can be freely transmitted.

Here’s how it goes in reality. Each institution trains a local model on its own private data, which remains behind the institution’s firewall. The institution then transmits only the model’s learnt parameters (weights and biases) to a central server. The server aggregates the parameters from all participating institutions to create an updated global model, which is then sent back to all institutions. This technique is repeated in numerous cycles. No patient record leaves its original institution (Pati et al., 2024).

The global federated training objective is to minimise a weighted combination of each client’s loss on its own data:

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{N} F_k(w)$$

where $F_k(w)$ is the local loss of client k , n_k is the number of training examples at client k , and $N = \sum_k n_k$ is the total training data across all clients. The issue is to reduce this global target without revealing any client’s local data (Nguyen et al., 2022).

2.7 Federated Averaging (FedAvg)

FedAvg is the foundational federated optimisation algorithm and the benchmark against which all federated approaches are measured. Each communication round has four steps:

1. The server sends the current global model to all participating clients.
2. Each client trains the received model on its own local data across numerous epochs.
3. Each client communicates its modified model parameters to the server.
4. The server uses weighted averaging to incorporate all received parameters into a new global model.

$$\theta_{\text{global}} \leftarrow \sum_{k=1}^K \frac{n_k}{N} \theta_k$$

The weight matrix and bias vector of each layer are aggregated independently. Larger clients, those with more training data, contribute proportionally more to the overall model.

The primary practical advantage of allowing several local training phases prior to each aggregate improves communication efficiency. McMahan et al. (2017) found that this strategy required 10 to 100 times fewer communication rounds than simply average gradients after each local step. In reality, fewer rounds lead to less network overhead and faster convergence.

An important limitation: FedAvg works for logistic regression and MLP because their parameters are real-valued matrices that can be meaningfully averaged. Tree-based models like as Decision Trees, Random Forest, Gradient Boosting, and XGBoost cannot be federated using parameter averaging since their architecture (branching rules, feature splits, thresholds) cannot be integrated by taking an average (McMahan et al., 2017).

2.8 FedProx

FedProx was presented by Li et al. (2020) to solve a weakness of FedAvg that becomes evident when data distributions differ between clients, which is virtually invariably true in real healthcare settings.

The problem is known as client drift. FedAvg allows each client to optimise its local target separately for several steps before providing its parameters to the server. When clients have various data distributions, they are effectively optimising for multiple objectives. After numerous local steps, their models have strayed greatly from one another. When the server attempts to average these disparate models, the outcome may not be favourable to any client’s real data distribution. Under sufficiently high heterogeneity, this drift can substantially compromise global model quality (Li et al., 2022).

FedProx addresses this by incorporating a proximal term into each client’s local training target. As training progresses, this term functions as an elastic leash, pulling each client’s model back towards the current global model.

$$\min_{w_k} \left[F_k(w_k) + \frac{\mu}{2} \|w_k - \theta_{\text{global}}\|^2 \right]$$

In gradient terms, each local update step becomes:

$$\text{gradient} = \text{local gradient} + \mu(w_k - \theta_{\text{global}})$$

The parameter μ controls how tight the leash is pulled. When $\mu = 0$, FedProx reduces to FedAvg. Clients’ ability to deviate from the global model becomes more limited as μ rises. Crucially, the proximal term is computed entirely locally, resulting in no additional communication cost. The parameters exchanged between clients and server are the same size as regular FedAvg.

Asad et al. (2026) and Mathew and Asha (2024) revealed that FedProx outperforms FedAvg on non-IID clinical tabular data, providing empirical support for its inclusion in this thesis.

2.9 Non-IID Data in Federated Learning

The Heterogeneity Problem

The acronym IID stands for independently and identically distributed, a statistical assumption that each client’s data is a representative random sample from the same global distribution. In reality, this is rarely the case in healthcare. distinct hospitals serve distinct patient populations. A hospital in the city center may see a younger, more diversified population of patients. A rural clinic could serve an older, more similar community. A specialist diabetic center encounters a significantly different

case mix than a normal practice. Because of these demographic and prevalence disparities, data distributions among clients are truly diverse.

This is significant theoretically since FedAvg’s theoretical convergence guarantees the assumption of IID circumstances. Under non-IID conditions, the algorithm may act irrationally, converge slowly, or converge to a global model that does not serve any individual client especially well. On medical datasets, Babar et al. (2024) found accuracy decreases ranging from 5 to 15 percentage points when non-IID circumstances were used instead of IID. Milasheuski et al. (2024) shown that the impact of heterogeneity differs significantly across federated algorithms on tabular clinical data.

IID Setting

In an IID scenario, each client receives data sampled evenly from the global distribution, resulting in similar class proportions and feature distributions. In this idealised situation, all hospitals would have statistically identical patient populations. This is the most beneficial situation for federated learning and is used as the experiment’s baseline distribution.

Non-IID via Dirichlet Distribution

Controlled experiments cannot be based on real-world institutional differences. Instead, non-IID situations are simulated using the Dirichlet distribution, which has become the community-standard way across the federated learning literature—validated across over 235 federated learning studies (G et al., 2024).

The class distribution among clients is derived from:

$$\pi \sim \text{Dirichlet}(\alpha, \alpha, \dots, \alpha)$$

The concentration value (α) is the important parameter. Consider it as managing how similar the customers are to each other:

- Large α (e.g., 100): All clients have approximately equal class proportions close to IID.
- $\alpha = 1.0$: **Mild heterogeneity** - clients differ visibly but not significantly.
- $\alpha = 0.5$: **Moderate heterogeneity** - indicates significant distributional differences amongst clients.
- $\alpha = 0.1$: **Strong heterogeneity** - certain clients may have a high proportion of diabetic patients, while others may have relatively few, closely resembling genuine disparities across hospitals serving quite diverse regions.

Client Drift

Under non-IID situations, each client’s local training gradient represents its own specific data distribution. After numerous local gradient steps, the models at various clients moved in directions shaped by their respective data, pulling them apart and further from the global optimum. When the server averages these disparate models,

it combines updates that were individually optimised for a distinct aim. The resulting global model may not be particularly effective in meeting any client’s objectives. This is the client drift issue, which is directly responsible for FedProx’s proximal penalty term (Li et al., 2020; Li et al., 2022).

2.10 Communication Cost

Every round of federated training requires the server to communicate the current global model to all clients and each client to return its updated model. Over several rounds and numerous clients, this bidirectional parameter exchange builds into a large communication overhead a genuine deployment cost that must be quantified honestly rather than ignored (Nguyen et al., 2022).

The total communication cost during training is computed as:

$$\text{params_bytes} = P \times B$$

$$\text{round_bytes} = \text{params_bytes} \times \text{clients} \times 2$$

$$\text{cumulative_bytes} = \sum_{r=1}^R \text{round_bytes}_r$$

where P is the number of model parameters, B is the size of each parameter (8 bytes for 64-bit double precision), and R is the number of training rounds. The factor of two reflects the bidirectional nature of each round, in which the server sends the model down and each client sends it back up.

The model architecture has a direct impact on communication costs. Logistic regression with d features has only $d + 1$ parameters (one per feature plus a bias), making it incredibly cheap to transmit. The shallow MLP has $d \times h + 2h + 1$ parameters that increase with input size and hidden layer width, making each round much more expensive.

An important concept is the convergence round, which occurs when model performance stabilises and further training yields no meaningful progress. Beyond this limit, the system spends communication resources with no return on model quality. Identifying the convergence point determines the optimal operating point for a federated deployment.

It’s important to understand what federated learning does and doesn’t achieve in terms of communication. It does not lower total data transfer compared to isolated local training; rather, it introduces iterative overhead that would not exist otherwise. It replaces the legally banned one-time communication of huge volumes of raw patient records with the legally permitted iterative transmission of small amounts of

model parameters (McMahan et al., 2017; Nguyen et al., 2022).

2.11 Probability Calibration

The Calibration Problem

A probability model makes a claim, such as "I believe there is a 70% chance this patient has diabetes." Calibration examines whether that assertion is valid in the real world. A correctly calibrated model satisfies the following: of all patients allocated a 70% risk, exactly 70% have diabetes. Formally:

$$P(Y = 1 \mid f(x) = p) = p \quad \forall p \in [0, 1]$$

The difficulty is that most current classifiers, particularly ensemble approaches like Random Forest and Gradient Boosting, as well as neural networks, are consistently miscalibrated. They don't distribute their estimated probability evenly over the range. Instead, probabilities cluster toward the extremes: patients are assigned risks near 0% or near 100% significantly more frequently than the evidence suggests. The model is overconfident; it appears to be more certain than it is (Guo et al., 2017).

Overconfidence is unhelpful in general prediction tasks. In a clinical setting, it is a true patient safety concern. Healthcare experts make treatment decisions based on risk levels, not merely whether the model says "positive" or "negative." A patient who is 85% likely to develop diabetes may be prioritised for more severe action, but a patient who is 40% likely to get diabetes may receive less frequent monitoring. If those probabilities are not calibrated—if the 85% patients are actually 60% and the 40% patients are truly 55% resource allocation, treatment intensity, and clinical priority are all altered (Jagatheesaperumal et al., 2024).

Calibration has mostly been lacking from the federated diabetes prediction literature. No existing federated diabetes study employs post-hoc calibration of predicted probabilities (Islam et al., 2022; Tang et al., 2024; Su et al., 2023; Gawande et al., 2025). One of the thesis' main contributions is to fill this gap.

Reliability Diagram

The reliability diagram is the common representation of calibration quality. The anticipated probabilities (0–1) are divided into M equal-width bins. For each bin, two values are generated and shown against one another: the mean projected probability (what the model claims) and the observed fraction of positive outcomes. A correctly calibrated model is exactly on the diagonal; predicted risk equals actual risk. Points below the diagonal indicate overconfidence (the model claims greater certainty than the results justify). The points above suggest a lack of confidence (van Calster et al., 2020).

Brier Score

The Brier Score is the mean squared difference between expected probability and genuine binary outcomes.

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2$$

The lower the better. A model that assigns 0.9 to a non-diabetic patient suffers a significant penalty (0.81). A model that assigns 0.5 to the same patient pays a significantly lower penalty (0.25). The Brier Score penalises both bad discrimination and poor calibration, making it a more comprehensive measure of probabilistic prediction quality (van Calster et al., 2020).

Expected Calibration Error (ECE)

The main calibration metric used in this thesis is ECE. It calculates the average difference between observed and predicted rates, weighted by the number of patients in each bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$$

The model is correctly calibrated if its ECE is 0. Greater systematic differences between stated confidence and actual accuracy are indicated by higher values. ECE is susceptible to miscalibration not only at the extremes but throughout the entire probability range (Silva Filho et al., 2023).

Log Loss

Log Loss penalises confident inaccurate predictions heavily.

$$\text{Log Loss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

Lower is preferable. A model that confidently assigns a probability of 0.99 to a non-diabetic patient pays a log loss of approximately 4.6 for that prediction. A model that assigns 0.6 to the same patient pays only about 0.92. Log Loss rewards models that are not only accurate, but also properly humble when faced with uncertainty (Silva Filho et al., 2023).

Uncalibrated Baseline

When no calibration is performed, raw model probabilities are used directly. This serves as the standard against which calibrated versions are measured. For ensemble models and neural networks, uncalibrated probabilities are typically clustered near 0 and 1, visible as points hugging below the diagonal in reliability diagrams (Guo et

al., 2017; Odesola et al., 2025).

Sigmoid Calibration (Platt Scaling)

Sigmoid calibration, also known as Platt scaling, learns a simple two-parameter transformation of the model’s raw output scores. It applies a logistic function to a held-out calibration dataset.

$$p_{\text{calibrated}} = \frac{1}{1 + e^{-(a \cdot p_{\text{raw}} + b)}}$$

The parameters a and b are determined by reducing cross-entropy in calibration data. Initially, it expands or compresses the probability scale, shifting it up or down. Sigmoid calibration works well when the miscalibration pattern is roughly S-shaped, which is common for logistic regression. It may not be sufficient for ensemble methods and neural networks, which have more complex miscalibration patterns (Ojeda et al., 2023).

Isotonic Regression Calibration

Isotonic regression calibration offers more flexibility than Platt scaling. Rather of assuming any specific functional form, it fits a completely general static non-decreasing function to the calibration data.

$$p_{\text{calibrated}} = g(p_{\text{raw}})$$

$$\min_g \sum_{i=1}^m (y_i - g(s_i))^2 \quad \text{subject to} \quad g(s_i) \leq g(s_j) \text{ when } s_i \leq s_j$$

The asymmetry constraint is significant because it ensures that if the model ranks patient A as higher risk than patient B, the calibrated probabilities will maintain that ranking. Platt scaling cannot represent complex non-linear miscalibration patterns, although isotonic regression can. Ojeda et al. (2023) demonstrated that it consistently outperforms sigmoid calibration for non-sigmoidal miscalibration, whereas Odesola et al. (2025) verified this for ensemble models using structured clinical data.

Global Calibration

Global calibration fits a single calibration function to the entire shared calibration dataset, which includes all patients and institutions combined. It has the most data and achieves the best statistically stable calibration fit. Both sigmoid and isotonic algorithms can be used in global mode. Global calibration indicates the calibration performance limit, or what is possible when calibration data can be centralised (Majlatow et al., 2025).

Federated Calibration

Calibration data cannot be centralised in a federated context since it is subject to the same legal constraints as training data. After the global model has converged, each client separately applies its own calibration function to its own local calibration partition. For federated sigmoid, each client c learns its own local parameters:

$$p_{\text{calibrated},c}(s) = \frac{1}{1 + e^{-(a_c s + b_c)}}$$

In federated isotonic, each client fits its own sequential function g_c . When a new patient requires a forecast, the calibrated probability is the mean of all client calibrators:

$$p_{\text{calibrated}}(s) = \frac{1}{K} \sum_{c=1}^K \text{calibrator}_c(s)$$

This approach ensures data locality throughout the pipeline. Patient data never leaves the originating institution during model training or calibration (Majlatow et al., 2025).

2.12 Decision Threshold Selection

After establishing calibrated probability, a decision threshold transforms these into binary clinical decisions. Two strategies are being evaluated.

Fixed threshold ($t = 0.5$): A patient with a forecasted risk of 50% or above is considered positive. This is the most common default and is simple to understand, but it is not always the appropriate threshold for a certain clinical situation or dataset imbalance.

F1-optimal threshold: Instead of utilising a set value, the threshold is chosen by determining the value that maximises the F1 score on the calibration split:

$$t^* = \arg \max_{t \in [0,1]} F1(t) \text{ on calibration data}$$

This threshold is then applied unchanged to the test set. The key is that threshold optimisation employs only calibration data, not the test set. Using test data for threshold selection would result in evaluation leakage, biasing the reported metrics optimistically (Huang et al., 2020).

2.13 Evaluation Metrics

All classification performance metrics are derived from four fundamental counts comparing the model’s predictions $\hat{y}$ against the true labels y_{true} on the test set: **True Positives (TP)** diabetic patients correctly identified; **True Negatives (TN)** non-diabetic patients correctly identified; **False Positives (FP)** healthy patients

wrongly flagged; and **False Negatives (FN)** diabetic patients wrongly cleared.

Accuracy is the fraction of all patients classified correctly.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy is straightforward, yet it can be extremely misleading on imbalanced datasets. If just 10% of patients have diabetes, a model that classifies everyone as non-diabetic achieves 90% accuracy but is medically ineffective. It is recorded, but not used as the key statistic (Chang et al., 2023).

Precision evaluates the reliability of the model’s positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision implies that when the model identifies someone as diabetic, it is typically correct.

Recall (Sensitivity) evaluates the proportion of real diabetes patients caught:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall means that the model detects the majority of diabetic patients, but at the risk of producing more false alarms.

Specificity determines what percentage of healthy individuals are successfully treated:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Balanced Accuracy compares performance across both classes, making it more suited than raw accuracy for imbalanced datasets:

$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2}$$

F1 Score represents the harmonic mean of precision and recall. It rewards models that are both precise and comprehensive, avoid burdening physicians with false positives or missing patients that require care:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 is the key classification quality statistic used throughout this thesis (Kaliappan et al., 2024).

The AUROC (Area Under the Receiver Operating Characteristic Curve) is the likelihood that the model will assign a greater risk score to the diabetic patient

if one diabetes patient and one healthy patient are picked at random. It varies between 0.5 (random guessing) to 1.0 (perfect discernment). Crucially, it is threshold-independent, meaning it assesses the validity of the probability ranking across all thresholds. This makes it the fundamental indicator for evaluating federated versus centralised performance, with the purpose of determining how well the model discriminates independent of the operational point used (Silva Filho et al., 2023).

PR AUC (Precision-Recall Area Under Curve) is the area under the curve that shows precision versus recall at all thresholds.

$$AP = \sum_k (\text{Recall}_k - \text{Recall}_{k-1}) \times \text{Precision}_k$$

PR AUC is especially useful when the positive class (diabetes patients) is small, as it focuses solely on how well the model handles positive instances rather than weighting performance by the huge number of true negatives (Kaliappan et al., 2024).

2.14 Explainable AI - SHAP

A model that states "this patient is 82% likely to develop diabetes" without explaining why is difficult to apply properly in a clinical environment. Healthcare professionals must grasp the reasoning behind forecasts in order to trust them, determine when they may be incorrect, and communicate effectively with patients about their risk factors (Awosika et al., 2024).

This is more than just a practical matter; it is also a legal one. GDPR Article 22 establishes a legal right to an explanation for automated choices that have a significant impact on humans, as clinical AI systems clearly do. Explainability is a regulatory necessity in Europe, where this topic is set, rather than an optional extra.

The federated setting adds an extra layer of complexity. Ducange et al. (2024) showed that in non-IID conditions, SHAP feature significance levels can vary significantly across clients. When the same global model is applied to two hospitals with distinct patient populations, the risk of diabetes may be assigned to various characteristics at each institution. This presents an important clinical governance question: if explanations differ across institution, which one should drive clinical practice? This thesis addresses this issue by comparing explanation stability over rounds and explanation agreement among clients.

SHAP Values

Lundberg and Lee (2017) developed SHAP (SHapley Additive ExPlanations) as a systematic methodology for analysing any model's prediction into contributions from specific characteristics. The approach is based on cooperative game theory, with each characteristic viewed as a player in a game, and its contribution calculated as the average marginal impact across all potential ways it could be included or deleted

from the forecast.

SHAP breaks down a single prediction, $f(x)$, as follows:

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j$$

The baseline prediction (ϕ_0) is the average prediction across all patients, while ϕ_j represents the contribution of feature j to this specific prediction. Positive ϕ_j levels indicate a higher likelihood of diabetes, while negative values suggest opposite.

For linear models, the contribution simplifies to a straightforward formula: each feature's contribution is simply the distance it deviates from the baseline, scaled by the model's learnt weight. For non-linear models, it is approximated by calculating how much the prediction changes when feature j is replaced with its baseline. SHAP ensures three critical properties: local accuracy (the contributions sum exactly to the prediction), missingness (features not in the input receive zero contribution), and consistency (if a feature becomes more important in the model, its SHAP value cannot decrease).

Global Feature Importance

SHAP data are aggregated to determine which attributes are most important throughout the patient group overall.

$$\text{Mean Absolute SHAP}_j = \frac{1}{n} \sum_{i=1}^n \left| \phi_j^{(i)} \right|$$

Features with higher mean absolute SHAP values have more consistent influence. This global rating enables physicians to determine whether the model's priorities are consistent with accepted medical knowledge regarding diabetes risk factors. For example, does the model appropriately prioritise glucose level over, indicate a less clinically important feature? Islam et al. (2025); Mohanty et al. (2024).

Explanation Stability

A practically effective model should converge in both performance and explanations during a federated training procedure that spans multiple rounds. Even though the AUROC appears to be fine, the clinical interpretation of the model is unstable if the model's feature priority ranking varies significantly from round to round.

Spearman rank correlation between feature priority rankings in successive rounds is used to gauge explanation stability:

$$\rho = 1 - \frac{6 \sum_j d_j^2}{d(d^2 - 1)}$$

where the rank difference of feature j between two successive rounds is represented by d_j . A score of 1.0 indicates "perfect stability," or no fluctuation in the ranking.

Rankings are largely random if the value is close to 0. High stability means that in addition to its prediction performance, the model has learned a consistent clinical narrative (Ducange et al., 2024).

Cross-Client Feature Agreement

Even after training is standardised globally, different clients may have learnt to weight aspects differently based on their local patient populations. Top-k overlap quantifies the extent to which two customers agree on the most significant attributes.

$$\text{Top-k Overlap} = \frac{|\text{TopK}_A \cap \text{TopK}_B|}{k}$$

A number of 1.0 indicates that two clients agree totally on the top k qualities. A number of 0.4 indicates that 40% of their top k traits overlap. Under strong non-IID conditions, decreased overlap is expected, as different hospitals see distinct clinical trends. Understanding the scope of this difference has practical consequences for how federated model explanations should be conveyed and used by member institutions (Ducange et al., 2024).

2.15 Algorithmic Fairness

A model with 85% overall accuracy can still be systemically biased if it performs well for some patient groups but poorly for others. Concerns about fairness are especially acute when predicting diabetes. Diabetes is more common among older patients and those with a higher BMI; this is clinical reality, not discrimination. However, systematic disparities in error rates between groups missing diabetic patients at a higher rate in one demographic group than another, or falsely flagging healthy patients at different rates constitute true injustice that can cause real suffering (Xu et al.).

The federated setting is important from a fairness standpoint. According to Chen et al. (2023), training on population varied datasets, as achieved by multi-institutional federation, can serve as a bias mitigation strategy as compared to training on homogenous data from a single institution. Pfohl et al. (2021) shown, however, that the fairness-accuracy trade-offs must be measured empirically rather than assumed. This thesis evaluates fairness across age groups and BMI categories for all models and training paradigms.

Demographic Parity Difference (DPD)

Demographic parity investigates whether positive predictions are provided at equal rates across various groups. The selection rate for group g is the proportion of that group predicted positively. The Demographic Parity Difference is the difference between the greatest and lowest selection rates.

$$\text{DPD} = \max_g P(\hat{y} = 1 \mid G = g) - \min_g P(\hat{y} = 1 \mid G = g)$$

A DPD of 0 indicates that all groups are projected positively at the same rate. A greater score suggests that some groups are flagged more frequently than others. The weakness of DPD is that it assumes identical prediction rates are completely desirable, even when true prevalence disparities exist between groups. Even if a model accurately reflects that older patients have greater diabetes rates, it will still display a non-zero DPD (Xu et al., 2022).

Equalized Odds Difference (EOD)

Equalised Odds is a more clinically relevant fairness criteria. Rather than requiring equal prediction rates, it demands that the model’s error rates, including false positive and true positive rates, be equal across groups. The Equalised Odds difference is:

$$\text{EOD} = \max \left(\max_{g,g'} |TPR_g - TPR_{g'}|, \max_{g,g'} |FPR_g - FPR_{g'}| \right)$$

This indicates that the model should catch all patients with diabetes at the same rate, regardless of demographic category. Furthermore, the model should generate false alarms at the same rate across all healthy patients, independent of group. An EOD of zero represents perfect equalised odds.

Equalised odds have clinical appeal because they allow for actual prevalence differences across groups while requiring equal error treatment. If older patients have a higher diabetes prevalence, a fair model can predict positive more often for older patients. However, it should not miss diabetic older patients at a higher rate than diabetic younger patients, or raise false alarms for healthy older patients at a different rate than healthy younger patients. Both Xu et al. (2022) and Chen et al. (2023) argue that equalised odds are the better fairness requirement for clinical prediction tasks. It is the fundamental fairness metric in this thesis.

Chapter 3

Related Work and Literature Review

3.1 Federated Learning in Healthcare and Diabetes Prediction

Foundational Work and Healthcare Surveys

McMahan et al. (2017) formally introduced the federated learning paradigm with their Federated Averaging algorithm, which offered iterative model parameter averaging across decentralized clients as a more communication-efficient alternative to synchronous gradient descent. Their research exhibited 10 to 100 times fewer communication rounds across five model architectures and four datasets, proving the technical viability of distributed collaborative training. However, McMahan et al. did not address healthcare applications, probability calibration, or model interpretability, and their convergence guarantees are based on IID data distributions that do not apply to realistic multi-institutional clinical deployments. Despite these drawbacks, FedAvg remains the universal baseline algorithm in federated learning research, and it serves as the foundation for this thesis’s federated training pipeline.

As federated learning (FL) evolved into a more general-purpose paradigm, healthcare emerged as one of its most important application domains. Nguyen et al. (2022) conducted a comprehensive assessment of FL in smart healthcare, including resource-aware, safe, privacy-preserving, and personalized FL applications. Their research identified several ongoing issues, including statistical heterogeneity, communication overhead, system heterogeneity, privacy attacks, and a lack of interpretability tools, which directly inspired the SHAP-based explainability component of this thesis. Teo et al. (2024) conducted a systematic analysis of 612 healthcare FL papers and showed the increasing expansion of healthcare FL research while noting that organized tabular clinical data remains neglected when compared to medical imaging. This gap emphasizes the need to assess FL using the CDC tabular dataset used in this investigation. Similarly, Pati et al. (2024) pointed out that building trust in federated healthcare AI necessitates not just good predictive performance but also interpretable and transparent models to facilitate clinical adoption. Despite addressing privacy, trust, and explainability, these surveys do not look at probability calibration, which is an important aspect of model trustworthiness that this thesis aims to address.

Pan et al. (2024) suggested an adaptive FL framework for clinical risk prediction utilizing multi-hospital EHR data that addresses distribution drift by dynamically

adjusting aggregating weights over training rounds rather than using traditional FedAvg’s fixed dataset-size weights. Their framework outperformed static FedAvg on EHR-based clinical outcomes, suggesting that performance-aware aggregation is useful in diverse clinical situations. This discovery is conceptually consistent with the softmax-weighted aggregation technique used in this thesis, in which clients with greater local validation performance contribute proportionally more to global model updates. However, Pan et al. do not use explainability or calibration in their predictions, and their emphasis on administrative EHR outcomes rather than binary disease classification limits direct comparison with the diabetes prediction task investigated in this thesis.

Federated Learning Applied Directly to Diabetes

Islam et al. (2022) provide the most relevant real-world justification for this thesis in the federated diabetic literature. Their study described a decentralized FL system that predicts diabetes-related complications by leveraging real clinical EHR data from several healthcare organizations. Using FedAvg with class balancing strategies to overcome dataset imbalance, their federated models outperformed centralized learning, proving that privacy-preserving collaborative diabetic modelling is achievable on real multi-site clinical data. However, Islam et al.’s evaluation is completely focused on accuracy and class imbalance metrics. The work does not analyze whether projected probabilities are well-calibrated, does not provide feature-level explanations for model decisions, and does not compare performance across non-IID heterogeneity levels. These omissions collectively form the fundamental gap that this thesis addresses.

Tang et al. (2024) demonstrated one of the first real-world implementations of federated learning (FL) for diabetes prediction, utilizing cross-province primary care data from the Canadian Primary Care Sentinel Surveillance Network. Their findings showed that a federated multilayer perceptron (MLP) performed similarly to, or slightly better than, its centralised version, however federated logistic regression performed worse. This diversity in model behaviour throughout federated training emphasizes the significance of comparing several machine learning and deep learning models, which directly drives the comparison strategy used in this thesis. However, the study concentrated on predicted performance indicators and did not examine model calibration or explainability. Because clinical decision-making requires both solid probability estimations and accurate forecasts, the absence of calibration analysis limits the framework’s practical application.

Su et al. (2023) addressed both privacy and security in a multi-party FL system for diabetes risk prediction, creating customised safe protocols for several model families such as XGBoost, LightGBM, neural networks, and logistic regression on Seceum FL platform. Their findings showed that federated models using secure computing may match centralised prediction performance while protecting against gradient leakage attacks, representing a significant advancement in the security dimension of federated healthcare AI. However, Su et al. do not discuss probability calibration, SHAP-based explainability, or non-IID robustness. Their evaluation framework focuses solely on

predictive accuracy, leaving the clinical credibility of anticipated probabilities completely unexplored.

Gawande et al. (2025) studied federated learning for diabetes prediction using the Flower framework and the Pima Indians Diabetes Dataset, which were shared across five simulated clients under non-IID conditions. The study trained a fully connected neural network and used LIME to assess model explainability. An interesting conclusion was that feature importance explanations differed amongst clients due to changes in local data distributions, emphasizing the role of data heterogeneity on interpretability. However, the study has a few shortcomings. It relied entirely on LIME and did not assess SHAP, which is extensively used for local and global model interpretation. Furthermore, the authors concentrated on prediction performance but did not evaluate probability calibration, leaving the reliability of estimated risk probabilities untested. Furthermore, the evaluation was limited to a single deep learning model, with no comparison to traditional machine learning approaches using federated training. As a result, the performance of various model types in federated diabetes prediction is uncertain.

Chellamani et al. (2025) investigated a FL framework for non-invasive blood glucose monitoring based on photoplethysmography signals from biosensor data, and achieved high clinical accuracy in Clarke error grid analysis. While their study illustrates FL’s usefulness in diabetes care, it focuses on continuous waveform prediction from physiological sensors rather than binary risk classification from structured tabular clinical data. The sensor-based modality, continuous output, and lack of XAI and calibration indicate that this project addresses a fundamentally different technical problem than this thesis.

Yang and Li (2025) used recurrent neural networks and continuous glucose monitoring data to create a clustering-based personalized FL model for glucose prediction. Their method increased prediction customisation by grouping comparable patients across clients before federated aggregation. While technically unique, their emphasis on sequential CGM data and continuous glucose output rather than structured tabular classification implies that their architecture and evaluation criteria are not directly comparable to this thesis’ binary diabetes risk prediction methodology. They do not include calibration or explainability in their evaluation.

FedEnTrust, a federated ensemble framework combining blockchain-based access control and knowledge processing for diabetes prediction on the Pima Indians dataset, was proposed by Hasan and Li (2026) in the most current paper in the evaluated literature. FedEnTrust, a 2026 publication that still relies solely on accuracy measures without calibration or XAI evaluation, stands out for achieving 84.2 percent accuracy. Practical implementation issues are raised by the blockchain overhead, which adds 210 milliseconds of latency per transaction. The relevance and significance of this thesis’s contribution are further supported by the fact that even the most current federated diabetic article under consideration leaves out calibration and explainability.

3.2 Non-IID Data Heterogeneity in Federated Learning

Benchmarks and Algorithmic Comparisons

The most thorough experimental benchmark of federated learning under non-IID conditions was carried out by Li et al. (2022), who evaluated FedAvg, FedProx, SCAFFOLD, and FedNova across nine datasets and proposed six non-IID partition strategies covering label distribution skew, feature distribution skew, and quantity skew. Their main conclusions have a major impact on this thesis. For all studied algorithms, non-IID situations result in a considerable reduction in accuracy. When the proximal parameter is tiny, FedProx exhibits little improvement over FedAvg, indicating that FedProx’s benefit is not always present. In all non-IID circumstances, no particular algorithm is dominant. Instead of assuming the superiority of one algorithm, these results support the empirical comparison of FedAvg and FedProx under both IID and Dirichlet-partitioned non-IID conditions in this thesis. They also support the evaluation of both configurations across various heterogeneity levels.

In a comprehensive IEEE Communications Surveys and Tutorials study of 235 federated learning articles, G et al. (2024) gave a full taxonomy of non-IID data sources and defined Dirichlet distribution partitioning as the community standard for modeling label skew in federated experiments. Their assessment indicated that research interest in non-IID FL peaked in 2023, and that FedProx and SCAFFOLD are the most widely used solutions, despite the fact that each handles only a subset of non-IID issues. This thesis’s non-IID simulation technique uses Dirichlet partitioning with three heterogeneity levels: mild ($\alpha = 1.0$), moderate ($\alpha = 0.5$), and strong ($\alpha = 0.1$).

Non-IID in Clinical Federated Settings

Babar et al. (2024) conducted an empirical study on FedAvg performance under IID and non-IID settings on a medical imaging dataset, discovering accuracy decreases of 5 to 15 percentage points with growing data heterogeneity. While their study employs imaging rather than tabular clinical data, their methodology of comparing IID and non-IID performance across different client counts and communication cycles serves as a direct prototype for the IID versus non-IID evaluation design in this thesis. Their measured performance degradation range also defines the magnitude of heterogeneity impact that federated diabetic architectures must manage.

Milasheuski et al. (2024) are especially notable in this thesis since they are the only study in the reviewed literature that evaluate FL algorithms using tabular clinical data — a stroke prediction dataset — rather than imaging or text data. After evaluating seven FL algorithms across different heterogeneity types, they discovered that SCAFFOLD works best under label skew and feature skew, whereas FedProx performs best under high heterogeneity for binary classification. Critically, they observed that the low dimensionality of tabular clinical information produces fundamentally different FL convergence dynamics than high-dimensional image data,

necessitating independent study. This finding clearly explains this thesis’s tabular-specific federated evaluation rather than generalizing from imaging FL outcomes, and it informs the inclusion of FedProx as a primary comparison to FedAvg.

Asad et al. (2026) performed a controlled simulation that is strikingly similar to this thesis, using FedProx to predict heart disease across four demographically stratified simulated hospital clients using 303 structured clinical patient records — a small tabular dataset comparable in scale to the Pima Indians diabetes setup. FedProx with $\mu = 0.05$ obtained 85 percent accuracy, beating both centralised training (83.33 percent) and the local-only average (78.45 percent) across 50 independent experimental runs. This finding is interesting because it shows that federated training with proximal regularisation can outperform centralized training on short clinical datasets, most likely because to the regularisation effect that prevents overfitting. However, Asad et al. address heart disease rather than diabetes and do not incorporate any XAI or calibration, leaving the reliability of their estimated probability untested.

Mathew and Asha (2024) found that using FedProx and FedSplit algorithm on medical IoT data under Dirichlet partitioning at three heterogeneity levels ($\alpha = 0.02, 0.05, \text{ and } 0.1$) improved test accuracy by 35% compared to standard FL under strong heterogeneity. Their use of Dirichlet partitioning and proximal regularisation directly informs the algorithmic architecture of this thesis, but their medical IoT focus, lack of a disease-specific prediction objective, and deletion of calibration and XAI make direct comparison difficult.

3.3 Probability Calibration in Clinical Machine Learning

Theoretical Foundations

Silva Filho et al. (2023) present a thorough examination of classifier calibration, laying the theoretical groundwork for evaluating and improving projected class probabilities. Their consideration of Expected Calibration Error (ECE), Brier Score, and Log Loss as complementary calibration measures informs the evaluation approach used in this thesis. The survey also highlights isotonic regression as a non-parametric, monotone-constrained calibration method that does not require a parametric shape assumption, making it ideal for heterogeneous federated data with varying miscalibration patterns among clients. Furthermore, the authors emphasize that well-calibrated classifiers accurately measure predictive uncertainty, which is essential for clinical decision-making based on diabetes risk scores.

Ojeda et al. (2023) investigated calibration methods with distributional disparities between training and validation populations. Their findings revealed that isotonic regression works well when there is enough calibration data, but it can overfit when the calibration dataset is minimal. This insight is especially relevant to the federated

setting explored in this thesis, since each client has access to a smaller local calibration set than would be available in a centralised system, which encourages the usage of cross-validation during calibration.

Calibration in Clinical Prediction

Huang et al. (2020) present a basic clinical informatics tutorial on calibration measures, demonstrating that prediction confidence is just as significant as prediction accuracy in clinical decision-making because treatment decisions are based on projected risk levels rather than binary labels. Their demonstration of reliability diagram building and calibration significance testing sets the standard clinical evaluation technique for this thesis.

Rajaraman et al. (2022) investigated calibration specifically under class imbalance in medical picture classification and discovered that calibration significantly improves model performance at the default 0.5 threshold under imbalance. Their discovery that isotonic regression is not advised for small datasets due to overfitting risk directly influences this thesis’s decision to use cross-validation during the client-level isotonic calibration step on the CDC dataset’s training partitions.

Odesola et al. (2025) performed a post-hoc calibration research using structured clinical heart disease data, which is one of the most methodologically relevant investigations in this thesis. When they compared six classifiers before and after Platt scaling and isotonic calibration, they discovered that isotonic regression consistently improved calibration performance, lowering Random Forest ECE from 0.051 to 0.011 and Brier Score from 0.007 to 0.002, whereas Platt scaling decreased KNN calibration. These findings lend significant empirical support to the adoption of isotonic regression as the principal calibration approach in this thesis. Furthermore, some of the classifier families tested by Odesola et al., such as Logistic Regression, Random Forest, and tree-based models, are included in this work, allowing for significant comparisons of calibration gains across investigations.

Koutsoubis et al. (2025) identified uncertainty quantification and calibration consistency across clients as key open difficulties in federated healthcare applications. Their identification of per-client calibration as a research gap directs the study of client-level calibration metrics, such as ECE and Brier Score, in this thesis. Majlatow et al. (2025) coupled isotonic regression calibration and SHAP-based explainability in a healthcare process monitoring scenario and discovered that calibration increased both the reliability of uncertainty estimations and the interpretability of SHAP explanations. This study reveals an important link between calibration quality and explainability, which motivates the suggested methodology to evaluate calibration performance and explanation stability concurrently.

Despite this set of calibration research, each federated diabetes study in the reviewed literature Islam et al. (2022), Tang et al. (2024), Su et al. (2023), Gawande et al. (2025), Chellamani et al. (2025), Yang and Li (2025), and Hasan and Li (2026) evaluates models solely on classification metrics, with no calibration assess-

ment. This universal exclusion across seven pooled diabetes investigations, despite the well-established clinical value of probability dependability, constitutes the thesis’s most obvious and directly actionable weakness.

3.4 Explainable AI for Federated Diabetes Prediction

Federated XAI Studies

Ducange et al. (2024) carried out one of the few experiments that combine federated learning and explainable AI in a hospital scenario. They used a federated MLP with SHAP explanations to forecast Parkinson’s disease development and examined model behavior in IID and multiple non-IID scenarios. Their findings revealed that SHAP explanations become less consistent among clients when data distributions diverge, underscoring the importance of data heterogeneity in interpretability. This discovery serves as an essential motive for the current thesis, which objectively assesses SHAP explanation stability throughout federated communication rounds. However, their study is focused on a distinct disease area and does not include probability calibration, limiting its relevance to calibrated diabetes risk prediction.

The most closely related study to this thesis is Kumar et al. (2025), which is the only one in the evaluated literature to combine federated learning, SHAP explainability, and isotonic regression calibration into a single framework. After isotonic calibration, they achieved 99% accuracy, 98% F1-score, a Brier Score of 0.01, and an ECE of 0.59 using Random Forest, Gradient Boosting, XGBoost, Logistic Regression, and Decision Tree models among five decentralized clients for liver disease prediction. Their SHAP study revealed that Direct Bilirubin, SGOT, and Alkaline Phosphatase were the most influential features. The methodological similarities with this thesis are significant, as both research use the same classical machine learning model families, isotonic regression calibration, and SHAP-based explainability.

Despite these similarities, significant discrepancies exist. Kumar et al. concentrated on liver disease prediction rather than diabetes, and they did not investigate the effect of IID versus non-IID data distributions on model performance or calibration. Furthermore, they did not examine SHAP explanation stability across federated rounds or fairness-related evaluation. The significant ECE seen despite isotonic calibration raises concerns regarding calibration effectiveness under federated settings, offering further motivation for the more comprehensive analysis conducted in this thesis across many levels of data heterogeneity.

XAI Applied to Diabetes in Centralised Settings

Several studies have used SHAP and LIME to predict diabetes in centralized settings, providing valuable clinical context for analyzing federated explanation findings.

Alam et al. (2025) used a 1D-CNN with SHAP to predict diabetes in women across two datasets, discovering that glucose was dominating in both groups, although BMI and pregnancy count had culturally dependent value. This cross-population variance in feature relevance drives this thesis’s assessment of SHAP stability under non-IID federated situations, where diverse client populations may exhibit similar variation. However, Alam et al. work in a centralized framework with no FL or probability calibration.

Tasin et al. (2023) integrated the Pima Indians Diabetes Dataset with private Bangladeshi clinical data and used XGBoost, SHAP, and LIME to assess model explainability. Their identification of glucose, BMI, age, insulin, and skin thickness as major diabetes predictors serves as a good clinical standard for validating the SHAP explanations presented in this thesis. However, their research did not include federated learning or probability calibration. Islam et al. (2025) used SHAP, LIME, and partial dependence analysis on diabetes prediction models, resulting in good predictive performance and thorough explainability. Their usage of GridSearchCV for hyperparameter optimisation serves as a methodological foundation for the local model tuning technique used in this thesis.

Hasan et al. (2025) used AutoML with different explainability techniques to predict diabetes and identified federated learning as a viable future direction for boosting generalization while protecting privacy. This recommendation supports the study’s current research direction. Mohanty et al. (2024) tested different machine learning models and discovered that glucose, BMI, and age were consistently the most important predictors of diabetes. Their findings provide additional support for the clinical usefulness and stability of SHAP-based feature significance analysis in diabetes prediction.

3.5 Fairness in Clinical Machine Learning and Federated Learning

Xu et al. (2022) conducted a thorough study of algorithmic fairness in computational medicine, identifying equalized odds — needing equal true positive and false positive rates across demographic categories — as the recommended fairness metric for clinical prediction tasks. Their reasoning is completely incorporated in this thesis: unlike demographic parity, which may incorrectly identify truly higher diabetes prevalence in older patients as unfair, equalized odds conditions on true disease status necessitate equal error behavior. Pfohl et al. (2021) found that optimising for fairness constraints during training reduces predictive performance within groups, which supports this thesis’s decision to report fairness metrics as transparency dimensions rather than training objectives. Chen et al. (2023) argued that federated learning can serve as a bias mitigation strategy by allowing for training across demographically diverse institutional datasets, which inspired this thesis’s comparison of subgroup fairness under IID versus non-IID federated conditions to determine

whether heterogeneous client distributions narrow or widen demographic disparities in error rates.

Despite the well-established importance of fairness evaluation in clinical AI established by these three foundational studies, none of the seven federated diabetes papers in the reviewed literature. Islam et al. (2022), Tang et al. (2024), Su et al. (2023), Gawande et al. (2025), Chellamani et al. (2025), Yang and Li (2025), and Hasan and Li (2026) evaluates demographic parity difference or equalized odds across any demographic subgroup. This widespread lack of fairness evaluation in federated diabetes research constitutes a clear and major need, which this thesis fills by incorporating subgroup fairness analysis across age and BMI categories as a required dimension of clinical readiness assessment.

3.6 Summary of Research Gaps

Reading across all reviewed studies, the following five distinct research gaps appear, which collectively support this thesis and set it apart from all previous work in this area.

The first gap is the total lack of probability calibration in any federated diabetes prediction study. Every federated diabetes publication assessed, including Islam et al. (2022), Tang et al. (2024), Su et al. (2023), Gawande et al. (2025), Chellamani et al. (2025), Yang and Li (2025), and Hasan and Li (2026), evaluates models solely on classification metrics and does not report Brier Score, ECE, or Log Loss. This absence remains despite many calibration articles in the literature confirming the need of trustworthy projected probabilities for clinical threshold-based decision-making (Silva Filho et al., 2023; Ojeda et al., 2023; Odesola et al., 2025).

The second gap in diabetes prediction is the lack of an integrated framework that combines federated learning, SHAP explainability, and probability calibration. Kumar et al. (2025) accomplish this integration for liver disease, making them the only study in the literature to include all three components—but not for diabetes, not on the CDC dataset, and not with systematic IID versus non-IID comparison or fairness evaluation.

The third gap is that SHAP explanation stability over federated communication rounds has never been fully assessed for diabetes prediction. Ducange et al. (2024) discovered explanation inconsistency as a concern for Parkinson’s illness in non-IID conditions, while Gawande et al. (2025) observed LIME variation across customers but did not quantify it. No study has assessed the Spearman rank correlation of SHAP feature importance across rounds as a formal federated stability metric for diabetes.

The fourth gap is the lack of a systematic comparison of traditional ML models to a shallow MLP for diabetes using federated training, calibration, and SHAP. Tang et

al. (2024) compare MLP to LR under FL, while Su et al. (2023) compare XGBoost, LightGBM, NN, and LR, but none use calibration or XAI, making a fair comparison of probability reliability impossible.

The fifth gap is the complete absence of subgroup fairness evaluation in combined diabetic trials. Despite the fact that Xu et al. (2022), Pfohl et al. (2021), and Chen et al. (2023) established fairness evaluation as a criterion for clinical ML deployment, none of the seven federated diabetic publications examine demographic parity or equalized odds across age or BMI subgroups.

This thesis addresses all five deficiencies at once by using a unified trustworthy method of inspection to assess federated diabetes prediction in terms of performance, calibration, explainability, fairness, and communication cost within a single reliable experimental pipeline.

4.1 Methodological Approach

This thesis uses a quantitative, experimental research approach since both research questions require answers that can only be obtained by conducting controlled experiments and measuring numbers, rather than opinions, interviews, or case studies. RQ1 inquires as to how closely a federated model's AUROC and F1 match those of a centralised model across various data distribution conditions, as well as the communication cost in bytes required to achieve that performance all three of which are quantities that exist only after training models under controlled conditions and measuring the outputs. RQ2 inquires as to how calibration and SHAP-based explainability affect measurable trustworthiness indicators such as ECE, Brier Score, Log Loss, SHAP rank stability across rounds, and equalized odds difference across subgroups each of which is a number derived from an experiment and cannot be argued just on the basis of theory.

Given this, the most appropriate study methodology was to create a single, consistent experimental pipeline through which every model, training paradigm, and configuration would travel identically. Rather than running a small number of disconnected experiments and writing about each individually, the entire grid of conditions — different models, different training setups, different client counts, different heterogeneity levels, and different calibration methods — was run through the exact same code path, with the exact same preprocessing, splits, and evaluation logic. This consistency is crucial for a thesis like this one because the entire argument is based on being able to say "federated AUROC is X and centralised AUROC is Y, and the only thing that differs between these two numbers is the training paradigm" — if even minor inconsistencies entered into preprocessing or evaluation between runs, the comparison would no longer be reliable. This chapter leads you through the pipeline step by step, in the order it was built and executed.

4.2 Problem Definition and Study Design

The primary prediction task in this thesis is expressed as binary classification. Given a patient's health indicators, the purpose is to determine whether the patient is diabetic (label 1) or non-diabetic (label 0). This is the identical task formulation used

by practically every paper assessed in the related work chapter.

To answer RQ1, the exact same models were trained in three completely different training paradigms: centralised, where one model sees all of the pooled training data; local-only, where each simulated client trains a model using only its own slice of data with no collaboration; and federated, where clients train locally but periodically share and combine model parameters via a central server. Training the same model family under all three paradigms, using the same underlying data split or pooled differently, is what allows us to directly measure the performance gap that RQ1 is referring to — without this three-way comparison, we would have no idea whether federated learning is "close to centralised" or not, because there would be nothing to compare it to.

To respond to RQ2, every federated run was performed with five different calibration circumstances — none, global sigmoid, global isotonic, federated sigmoid, and federated isotonic — and SHAP explanations were computed and logged at each communication round rather than simply at the conclusion. This was important because RQ2 is asking not just "is the final model calibrated," but also how calibration alters things and whether explanations are consistent as training progresses—neither of which can be answered by looking at one take at the conclusion of training.

The experimental design consists of three training paradigms (centralized, local-only, federated), two distribution conditions (IID, non-IID), two client counts (3, 5), three heterogeneity levels ($\alpha = 1.0, 0.5, 0.1$), six model families, and five calibration strategies. Every cell in this grid was run independently, and the comparisons required for both research questions — client versus federated, centralised versus federated, calibrated versus uncalibrated, IID versus non-IID, and ML versus DL — are simply different segments through the same grid of results rather than separate studies.

4.3 Dataset Description

The CDC Diabetes Health Indicators dataset was chosen for this thesis because it is taken from the Behavioral Risk Factor Surveillance System, an ongoing health survey conducted by the United States Centers for Disease Control and Prevention. This dataset was carefully chosen above the far more widely used Pima Indians Diabetes Dataset, and that decision must be clearly justified here.

The Pima dataset has three unique issues that make it an unsuitable choice for a federated learning simulation. First, it only has 768 total records, which appears manageable until you try to split it across 3 to 5 simulated hospital clients — each client then ends up with somewhere between 150 and 250 rows, which is barely enough data to train a stable model locally, let alone hold out a separate calibration partition on top of that. Second, it exclusively includes data from women of Pima Indian descent, which is a significant gender limitation that renders any fairness study across broader groupings worthless because the population is already relatively small.

Third, a number of its characteristics contain biologically impossible zero values — a glucose measurement of 0 or a blood pressure reading of 0 does not occur in a living person — implying that a significant chunk of the information must be imputed before it can be used.

The CDC dataset avoids all three concerns. It has a total of 253,680 records, which means that even when split across 5 federated clients, each client still has tens of thousands of rows to train on, providing a realistic approximation of what an actual hospital’s patient database may look like. The population also has a significantly greater demographic variation across age, gender, and BMI, which is exactly what is required to conduct a meaningful fairness analysis across those subgroups in Chapter 5. There were no biologically impossible zero values to fix.

Diabetes_binary is the target column in this dataset, with 0 indicating that the respondent does not have diabetes and 1 indicating that they do. This dataset has an uneven class balance, with 218,334 negative records and only 35,346 positive records, indicating that diabetes responders account for around 13.9 percent of the whole dataset. This imbalance was treated as a genuine feature of the real-world problem rather than something to artificially force into balance, because diabetes is a minority condition in the general population, and a model evaluation that ignores this reality by oversampling away the imbalance would not reflect how the model would actually behave when implemented.

4.4 Data Preprocessing

Before a single model saw a single row of data, each trial went through the same fixed preprocessing procedure. This consistency is important because if preprocessing was permitted to vary significantly between tests, any subsequent performance difference may not be safely attributed to the real variable being studied — it could just be a preprocessing effect.

Missing numeric values, if present, were filled up with the median value of that column obtained solely from the training partition. The median was chosen over the mean because health metrics such as BMI can have extreme outlier values, which would cause a mean-based fill to be misleading, whereas the median is more adaptable to those outliers. Continuous numeric features were then standardized using z-score scaling, which involved subtracting the training mean from each feature and dividing the result by the training standard deviation to produce a feature with approximately zero mean and unit variance. This phase is notably important for the gradient-based models in this thesis — logistic regression and the MLP — since these models update their weights using gradient descent, and features on wildly diverse numeric scales might cause certain weights to update much quicker than others, leading to unstable training. Even while tree-based models like Random Forest do not require scaled features, the same scaling was applied to all models equitably, ensuring that no model acquired an inappropriate preprocessing advantage over another. Categorical data

were processed using one-hot encoding, which generates a separate binary column for each category level rather than treating the categories in an artificial numerical order.

The source of the statistics is the most crucial factor in this entire preprocessing procedure. The median for imputation, as well as the mean and standard deviation for scaling, were determined solely from the training partition, and those fixed values were then applied to alter the calibration and test divisions; they were never recalculated individually for those partitions. This was done on purpose to avoid a subtle but serious type of data leakage: if, for example, the scaling statistics had been calculated from the entire dataset before splitting, information about the distribution of the test set would have quietly leaked backward into how the training data was prepared, making the model’s apparent performance appear slightly better than it is in a genuinely unseen-data scenario.

4.5 Data Splitting Strategy

Instead of the common two-way train/test split used in many machine learning projects, the entire dataset was divided into three fully independent, non-overlapping segments. The training partition contains 152,208 rows (60 percent of the data), and it is the sole partition from which any model can learn its parameters. The calibration partition has 50,736 rows (20 percent), and it serves only one purpose: to fit calibration functions and select decision thresholds. The test partition also contains 50,736 rows, representing the final 20%, and it is only handled once, at the very end of each experiment, to display the actual numbers.

The reason for using three partitions rather than two stems from a single risk: if the same data used to evaluate the model’s final performance was also used to tune its calibration or select its decision threshold, the resulting metrics would be optimistically biased in a way that is easy to overlook but seriously undermines the credibility of the results. For example, if you choose the threshold that produces the highest F1 score by explicitly comparing it to the test set, it will appear to be a wonderful threshold because you intentionally picked out the one that works best on that specific data. That is not a measure of how the model will perform on new, unseen patients; rather, it is a measure of how well you can fit a threshold to data for which you already have an answer. Since the entire argument behind RQ2 is based on calibration metrics being honest and trustworthy, keeping the calibration partition entirely separated from the test partition was viewed as a non-negotiable design decision throughout this thesis, rather than an optional best approach.

4.6 Federated Learning Environment Setup

To simulate a realistic multi-hospital federated environment without access to actual multi-institutional patient data, the training partition was divided into a number

of separate client partitions, each representing an independent simulated healthcare institution that trains its own local model and never transmits raw patient data anywhere. Throughout this thesis, two different client counts were tested — 3 clients and 5 clients — to see if the number of participating institutions has any meaningful effect on how close federated training comes to matching centralised performance, as well as how much it affects total communication cost, because more clients inherently means more parties exchanging parameters every round.

For the IID condition, the training data was partitioned so that each client had roughly the same proportion of diabetes and non-diabetic instances as the other clients. This depicts the best-case situation for federated learning, in which each participating hospital serves a similar patient population. This partitioning logic was implemented in the `partition_clients()` function of `data.py`, and it serves as the baseline distribution condition against which the more realistic non-IID condition is later compared.

4.7 Non-IID Simulation Design

In reality, real hospitals do not serve similar patient groups; for example, a hospital in a low-income neighborhood with limited access to healthy food options is likely to see a higher number of diabetic patients than a hospital in a wealthy location with a younger, more active population. To account for this type of realistic fluctuation in the federated experiments, the non-IID condition was simulated using Dirichlet distribution partitioning, which is the standard and most extensively used method in the federated learning research field.

In this thesis, three different concentration values, represented by α , were tested: $\alpha = 1.0$, which represents mild heterogeneity, where clients differ only slightly from one another; $\alpha = 0.5$, which represents moderate heterogeneity, where client differences become clearly noticeable; and $\alpha = 0.1$, which represents strong heterogeneity, where some clients may end up with very few diabetic cases at all while others have a much higher concentration. These three particular values were selected because, taken as a whole, they cover the realistic range that is frequently utilized in the literature on federated learning, ranging from a mild simulation of real-world variation to a worst-case test of the framework. Testing across the entire range, rather of just one heterogeneity level, allowed us to determine whether the federated architecture declines gracefully as conditions worsened, or if there is a certain point at which performance suddenly collapses. This non-IID partitioning logic was implemented in the `_dirichlet_partition()` method within `data.py`.

4.8 Model Selection and Implementation

Six different model families were employed throughout this thesis, but not every model could be utilized in every training environment – there was a purposeful tech-

nological constraint that determined which models appeared in which case.

The whole centralized training set was used to train six models for centralised training: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting Classifier, XGBoost, and the shallow MLP. Except for XGBoost, all of these were built using scikit-learn classes, including logistic regression, decision tree classifier, random forest classifier, gradient boosting classifier, and MLP classifier. XGBoost was implemented separately, using the specific xgboost library, because it is not part of scikit-learn and has its own, more complex training method.

For local-only training, four models—Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting—were trained independently at each client, with XGBoost and the MLP purposely excluded. The reasoning behind this is that the purpose of the local-only baseline is to provide a fair comparison point against federated training, so it made the most sense to use model families that could potentially be federated, rather than models that have no federated counterpart to compare against.

For federated training, only two models were employed in all experiments: Logistic Regression and shallow MLPs. The limitation is not random; it stems directly from how federated aggregation works. FedAvg and FedProx combine model parameters from each client using weighted averaging. This only makes mathematical sense for models with continuous numerical parameters, such as the weight matrices and bias vectors in logistic regression and neural networks. Tree-based models, such as Random Forest or Gradient Boosting, lack this property because a decision tree’s internal structure is a set of branching if-then rules and threshold values, and there is no reasonable way to "average" two different trees with different rules and shapes into one combined tree. Because of this fundamental constraint, tree-based models cannot be federated using traditional parameter averaging, hence they were completely excluded from the federated tests. Both the federated logistic regression and the federated MLP were implemented from scratch using NumPy rather than an existing federated learning library, as this provided full direct control over the exact gradient computation required to correctly implement the FedProx proximal penalty, as well as the independent parameter aggregation step on the server side.

4.9 Hyperparameter Configuration

FedProx’s proximal penalty strength was set to $= 0.01$ for all experiments using this approach. This particular value was chosen as a moderate setting, based on values reported in comparable federated healthcare studies reviewed in Chapter 3 — strong enough to meaningfully pull client models back toward the global model and reduce drift, but not so strong that it slows convergence to the point where the model never settles into a stable solution within the number of rounds tested.

For the federated MLP, the size of the hidden layer and the number of local training epochs conducted by each client every round were both set at constant levels

throughout all MLP experiments. This consistency was intentional: by keeping the architecture completely unchanged between runs, any difference in results when switching from 3 clients to 5 clients, or from IID to non-IID, can be confidently attributed to that specific variable being studied rather than the model secretly having a different shape or capacity in one run versus another.

4.10 Federated Aggregation Procedure

Every federated training run, whether using FedAvg or FedProx, has the exact same overall loop structure. At the beginning of a communication round, the server takes the current global model’s parameters and delivers an identical copy to each participating client. Each client then trains on its own local data for a set number of epochs, beginning with the received global parameters. Under FedAvg, this local training step is simply standard gradient descent on the client’s data. Under FedProx, the identical local training phase contains an extra proximal penalty term added to the gradient, which actively pushes the client’s parameters back toward the global model from which it began, preventing the local model from deviating too much during training.

Once each client has completed its local training for that round, it sends its revised parameters to the server. The server then uses a weighted average to combine every received set of client parameters, with each client’s contribution to the final average weighted in proportion to how much training data that particular client has—a client with more data has a proportionally larger say in the resulting global model. This freshly integrated model becomes the global model in the following round, and the entire process — distribute, train locally, send back, average — is repeated for the specified number of rounds. Furthermore, the current global model’s performance was checked and logged after each round, not just at the end of training, allowing us to plot and analyze the round-by-round convergence curves.

4.11 Probability Calibration Procedure

Whether centralized or federated, a model’s raw output probabilities were calibrated using the previously mentioned held-out calibration partition after training was complete. Under no circumstances was the test partition touched at this point. This thesis compares five different calibration conditions: using no calibration at all as the raw baseline, global Sigmoid calibration, global Isotonic calibration, federated Sigmoid calibration, and federated Isotonic calibration.

For the two global techniques, the procedure is simple: a single calibration function is fitted using the whole shared calibration partition at once, and that single fitted function is then applied uniformly to all predictions made on the test set. The technique for the two federated methods differs significantly and was intended

particularly to preserve the same privacy concept that drives federated learning in the first place. Rather than combining the calibration data in one location, each client fits its own unique local calibrator utilizing only its own private slice of the calibration partition. When a final calibrated probability is required for a certain forecast, the outputs from each client’s unique calibrator are simply averaged to produce the final result. It would have been technically easier to simply combine all the calibration data centrally after federated training was complete and calibrate everything at once, but doing so would have subtly reintroduced exactly the kind of centralized data exposure that federated learning is supposed to avoid in the very beginning, since calibration data is still real, genuine patient data that should never need to leave its originating client. This was a purposeful design choice rather than a temporary solution.

4.12 Decision Threshold Selection

After calibrating the probabilities, two distinct threshold procedures were evaluated to turn the continuous probabilities into final yes-or-no diabetes predictions. The first and simplest technique has a set threshold of 0.5; any patient expected to be at or over 50% risk is classed as diabetic. The second strategy searches the entire range of possible threshold values using the calibration partition, specifically identifies the single threshold that produces the highest F1 score on that calibration data, and then locks that exact threshold value in and applies it to the test partition without any further adjustment.

This second, F1-optimized technique was provided explicitly because of the imbalance in the underlying dataset. With diabetic cases accounting for approximately 14% of the data, a flat 0.5 threshold consistently caused almost every model to underestimate the positive class, missing diabetes patients significantly more frequently than would be acceptable in a real-world clinical setting. Running the full threshold filter highlighted this issue across the board, and using the F1-optimal threshold instead provided a noticeably more equitable basis for comparing models, as it prevented the comparison from being unfairly dominated by which model handled a single arbitrary cutoff value better.

4.13 Explainability Implementation

SHAP values were calculated for each federated and centralised model in this thesis to determine which specific health factors influenced individual diabetes forecasts. In the federated setting, this explanation step had to be carried out locally at each client, just like training and calibration — each client computed its own SHAP values by applying the final converged global model to its own private local data, because raw patient data would never be sent anywhere else solely to generate an explanation centrally.

To determine whether the model’s explanations remained consistent throughout training or shifted unpredictably from one round to the next, the ranked list of most important features produced at each round was compared directly against the ranked list from the previous round, using Spearman rank correlation to produce a single stability score for that round. Separately, to determine whether different clients were genuinely focusing on the same important features as each other rather than learning contradictory explanations, the top several most important features identified for each individual client were compared pairwise against every other client using a simple set-overlap calculation. Both of these explainability metrics were computed within `metrics.py` and saved alongside the regular performance and calibration metrics from the same runs.

4.14 Fairness Evaluation Procedure

Every model’s test set predictions were broken down and examined separately across three different subgroup categories — age group, BMI category, and gender to see if each model treated different types of patients consistently rather than always reporting a single overall number that may hide meaningful disparities. For each individual subgroup inside each of those three categories, three numbers were computed directly from the test predictions: the positive prediction rate, the real positive rate, and the subgroup-specific false positive rate.

Two summarized fairness indicators were then calculated using the subgroup-level data. The first, demographic parity difference, is simply the greatest difference in positive prediction rate between any two groupings being studied. The second, equalized odds difference, is the greatest difference in true positive or false positive rates between any two groupings. Both of these fairness metrics were calculated for each and every model and training configuration tested, not just for the configuration that produced the best overall performance this was intentional, because the whole point of including fairness as one of the five evaluation dimensions was to see how fairness behavior changes across conditions, not just to report one isolated number for the "best" model and stop.

4.15 Communication Cost Measurement

Every round of federated training recorded the precise number of bytes delivered in both directions between the central server and each participating client in order to provide an honest and accurate representation of the amount of data being transmitted. Since every value was stored using double-precision floating point representation, the number of trainable parameters in the model that was currently being trained—22 total parameters in the case of logistic regression, or however many weights and biases the MLP’s particular architecture produced in its case—was multiplied by 8 bytes per parameter. This figure was then further multiplied by the

number of clients sending and receiving parameters during that specific round.

These separate per-round byte counts were then added together across all rounds of a training run to give a single cumulative communication total for the whole run. With this cumulative figure available, it was possible to directly answer practical questions relevant to real-world deployment, such as how much more expensive it is in communication terms to run with 5 clients versus 3, or how much additional communication overhead the MLP carries when compared to logistic regression for a broadly similar level of predictive performance.

4.16 Evaluation Metrics Summary

The total set of measurements obtained from every single experiment in this thesis fits neatly into five separate groups, which correlate precisely to the five evaluation criteria that the entire study is based around. The performance group — accuracy, precision, recall, specificity, balanced accuracy, F1, AUROC, and PR AUC — was calculated using the necessary functions from scikit-learn’s metrics package. The calibration group — Brier Score, Expected Calibration Error, and Log Loss — was computed using a combination of scikit-learn’s built-in functions where available, as well as custom binning logic written specifically for the ECE calculation, because scikit-learn does not include an ECE function out of the box. The explainability group, which includes the mean absolute SHAP value per feature, round-to-round stability scores, and cross-client overlap scores, was calculated using the SHAP library’s explainer objects in conjunction with custom comparison logic written specifically for this thesis to handle the round-tracking and cross-client overlap calculations. The fairness group (selection rate, demographic parity difference, and equalized odds difference per subgroup) was calculated by applying proprietary subgroup-splitting logic directly on the saved test set predictions. Finally, the communication group bytes transmitted per round and cumulative total was computed directly from within the federated training loop while training was taking place, as this figure is entirely dependent on the model’s parameter count at the time of training rather than anything that could be calculated later from the final predictions.

4.17 Experimental Pipeline Summary

Bringing everything in this chapter together, every single experiment performed in this thesis followed the same end-to-end sequence of actions in the same precise order. Initially, the raw CDC dataset was loaded. Second, it was divided into the training, calibration, and test portions mentioned before. Third, the preprocessing statistics were fitted just for the training partition and then applied consistently across all three partitions. Fourth, the training partition was divided into the required number of simulated clients using either the IID or the non-IID Dirichlet option. Fifth, the model was trained using the paradigm being tested for that experiment —

centralised, local-only, or federated with FedAvg or FedProx. Sixth, the calibration partition was used to apply the calibration technique of choice. Seventh, the decision threshold was determined, either fixed or F1-optimized. Eighth, all performance and calibration metrics were compared to the undisturbed test partition. Ninth, SHAP explanations were created and their consistency was tracked over rounds, where applicable. Tenth, fairness metrics were calculated for the appropriate groupings. And, during the federated rounds, communication bytes were logged continually rather than only at the end. Running the identical nine-step pipeline repeatedly for each model, training paradigm, client count, alpha value, and calibration method examined in this thesis.

References

